Supplementary Information

Genome streamlining is pervasive across prokaryotic functional categories, but a distinctive cofactor-versus-resistance split links metal-gene investment to ecological specialisation

Heather R. MacGregor[1,2] and Adam P. Arkin[2,3]

[1]Department of Chemistry and Chemical Biology, University of California, Berkeley, CA 94720, USA
[2]Environmental Genomics and Systems Biology Division, Lawrence Berkeley National Laboratory, Berkeley, CA 94720, US
[3]Department of Bioengineering, University of California, Berkeley, CA 94720, USA

Contents

SM1. Gene list construction and evidence tiers

1.1 Candidate KO universe

The metal-gene list was assembled from three independent sources:

1. **KEGG ORTHOLOGY / BRITE hierarchies.** KOs classified under BRITE modules for heavy-metal resistance, metal transport, and metalloenzymes were downloaded in 2024. Only KOs with an explicit metal annotation in the KEGG reaction or definition field were retained.
2. **BacMet2 database** [1]. Experimentally validated metal resistance genes from BacMet v2.0. KEGG IDs were assigned by mapping BacMet gene names to KEGG gene identifiers via the KEGG REST API (`/find/genes/{name}` endpoint).
3. **FitnessBrowser** [2]. Genome-wide transposon fitness screens under metallic stress conditions. KOs with significant fitness defects ($|\text{fitness score}| > 1.5$, $|t| > 5$) under ≥ 1 metal stressor were included.

KOs appearing in ≥ 2 sources were cross-validated; those from only one source were retained as lower-confidence candidates with appropriate tier labels. The final list covers 730 KOs across 17 metals.

1.2 Evidence tiers

Each of the 730 candidate KOs was assigned to one of five evidence tiers:

Table S1. Evidence tier definitions.

Tier	Label	n KOs	Criterion
1	Tier 1	32	Multi-source validated: BacMet2 AND FitnessBrowser AND KEGG module
2	Tier 2	108	Clear KEGG module definition; one additional source (BacMet2 or Fitness)
2-Fitness	Tier 2-Fitness	116	FitnessBrowser-validated only; no KEGG module annotation
3-BacMet	Tier 3-BacMet	188	BacMet2 literature-curated; no fitness evidence
3	Tier 3	286	KEGG BRITE only; ambiguous or multi-function annotation

Primary KO set used in all confirmatory analyses: Tier 1 + Tier 2 = **140 KOs**. These have the strongest multi-source evidence and unambiguous KEGG functional annotations. The full list with curation rationale is provided in Supplementary Table S2.

1.3 Functional categories

Each KO was assigned a primary functional category based on KEGG annotation and BacMet2 gene function:

- **Transport / Homeostasis** — metal importers, exporters, metallochaperones, siderophore-uptake systems, and ferritins.
- **Resistance / Detoxification** — metal-efflux pumps, detoxification enzymes, and metal-responsive regulatory proteins that confer resistance.
- **Sensing / Regulation** — two-component sensor kinases, MerR-family metalloregulators, and other metal-sensing regulatory elements.

- **Cofactor Biosynthesis** — enzymes in Fe–S cluster assembly, molybdopterin (Moco) biosynthesis, and cobalamin pathway steps that incorporate metal cofactors.
- **Metal-dependent Metabolism** — metabolic enzymes whose catalytic activity requires a metal cofactor, supported by KEGG module assignment and fitness screen evidence.

Assignment is based on the most specific category for each KO; `overlap_flag = True` identifies six KOs with dual-function ambiguity in the full 730-KO list (not in the primary 140-KO set).

1.4 Metal coverage

The primary 140-KO set covers 17 metals: Fe, Zn, Co, Mn, Cu, Ni, S, Mo, Cd, Al, Tl, Ag, Hg, Au, Pb, As, and Bi. Metal annotations are listed per KO in Supplementary Table S2. Metal-specific PGLS subsets were defined by filtering to KOs whose `metals` field contains the target metal. Nine metals with ≥ 34 primary KOs were included in the metal-specific confirmatory analysis: Co (101 KOs), Fe (98), Ni (84), S (74), Cu (71), Zn (67), Tl (45), Al (34), Mn (34). For Al, Tl, and S — elements with fewer dedicated resistance systems — KOs were assigned based on fitness-screen evidence or documented biochemical function; full curation rationale for these three metals is provided in Supplementary Table S4.

1.5 Pfam / InterPro structural QC

To assess whether the primary KO set has independent structural evidence of metal interaction, we performed a three-step API pipeline for all 140 KOs:

1. **KO → gene:** KEGG REST `link/genes/ko:{KO}` filtered to representative model organisms in priority order (*E. coli*, *B. subtilis*, *P. aeruginosa*, *Synechocystis*, *M. tuberculosis*, *S. coelicolor*, *R. prowazekii*, *C. jejuni*, *B. pertussis*, *M. genitalium*); then `conv/uniprot/{gene_id}` to obtain a UniProt accession. Rate: 0.4 s per call; 3-retry exponential back-off.
2. **Gene → Pfam:** UniProt REST `/{acc}?fields=xref_pfam&format=tsv` to retrieve Pfam domain IDs.
3. **Pfam → clan:** InterPro API `/entry/pfam/{PF}/?format=json` → `metadata.set_info.accession` to identify the InterPro clan. Rate: 0.3 s per call.

Metal-binding domains were defined by five InterPro clans (verified in InterPro version 97+): CL0704 (HMA heavy-metal-associated domain), CL0344 (4Fe–4S ferredoxin), CL0486 (Fer2 / 2Fe–2S ferredoxin), CL0361 (C2H2 zinc finger), and CL0193 (MBB metal beta-barrel); plus ≈25 metal-binding singleton Pfams identified from literature.

Results: 10/140 (7.1%) KOs have ≥ 1 metal-binding clan or singleton Pfam. The 130 without clan evidence fall into three classes: `pfam_no_metal_clan` (113 KOs — Pfam domains found but none in metal-binding clans; predominantly ABC transporter ATPase / permease scaffolds PF00005 / PF00950 / PF01032, MerR-family sensors, and outer-membrane efflux proteins); `no_gene_found` (15 KOs — no representative bacterial gene in KEGG REST); and `no_pfam_domains` (2 KOs: ARN / K08197 and cbiL / K16915). The low InterPro coverage is expected: metal transporters act via substrate selectivity at the binding pocket rather than via catalytic metal-cofactor domains, and will systematically fail clan-based Pfam filtering (see also Fig. S14 and §5.1 of the main text).

SM2. Per-megabase KO density calculation

2.1 Genomic data sources

Primary dataset (bacteria, $n = 1{,}574$ genera): BERDL Spark cluster, namespace `kbase.ke_pangenome`. The `genus_clusters` table contains one row per GTDB genus with pangenome statistics (total gene count, mean genome size in bp, and the full KO annotation set across all genomes in the genus-level cluster). Phylum / kingdom annotations joined from `kbase.ke_pangenome.gtdb_taxonomy_r214v1`.

Geochemical replication dataset (soil concentration, $n = 482$ genera): `refdata.ngsa` (Australian National Geochemical Survey of Australia; Geoscience Australia). KO content extracted by the same pipeline, restricted to genera detected in Australian soil and sediment samples. Soil metal concentrations (ICP-MS, $\mathrm{mg\,kg^{-1}}$) joined by a 200 km radius spatial join between NGSA sampling sites and genus occurrences.

2.2 KO density formula

For each genus g:

$$\text{ko\_per\_mb\_primary}(g) = \frac{n_{\text{distinct primary KOs}}(g)}{\text{mean\_genome\_size\_Mb}(g)} \tag{S1}$$

where mean_genome_size_Mb = mean_genome_size_bp/10^6 averaged across all genomes in the genus cluster, and $n_{\text{distinct primary KOs}}$ counts unique KO IDs from the 140-KO primary set present in ≥ 1 genome of the genus.

2.3 Predictor standardisation

For PGLS, `ko_per_mb_primary` is z-scored within each analysis dataset:

$$\text{predictor\_z} = \frac{\text{ko\_per\_mb\_primary} - \mu(\text{ko\_per\_mb\_primary})}{\sigma(\text{ko\_per\_mb\_primary})} \tag{S2}$$

Standardisation is performed separately within each analysis dataset (bacteria, archaea, geochemical replication), so β values are not directly comparable across datasets.

2.4 Primary dataset summary

Table S2. Summary statistics for the primary bacterial dataset ($n = 1{,}574$ genera).

Statistic	Value
n genera	1,574
`ko_per_mb_primary`: mean (SD)	8.61 (3.83) KOs/Mb
`ko_per_mb_primary`: range	0.78–34.56 KOs/Mb
Mean genome size: mean (SD)	3.73 (1.63) Mb
Dominant phyla	Proteobacteria (43%), Firmicutes (21%), Actinobacteria (13%), Bacteroidetes (12%)

SM3. Ecological niche breadth quantification

3.1 Levins' standardised niche breadth

Niche breadth was quantified using Levins' B [3], standardised following Colwell & Futuyma [4]:

$$B = \frac{1}{\sum_i p_i^2} \qquad B_{\text{std}} = \frac{B-1}{n-1} \tag{S3}$$

where p_i is the proportional abundance of a genus in sample i (relative to total abundance of that genus across all samples) and n is the total number of samples. $B_{\text{std}} \in [0, 1]$: 0 = complete specialist (all occurrences in one sample), 1 = complete generalist (perfectly even distribution). Implementation: `scripts/niche_utils.py::levins_b_std()`.

3.2 Primary niche breadth dataset (MicrobeAtlas)

Data source: `arkinlab.microbeatlas` (BERDL Spark) — genus-level OTU abundance matrix across $>$ 400,000 global 16S rDNA amplicon samples aggregated in the MicrobeAtlas database [5]. Each OTU linked to a genus via `otu_pangenome_link_v2.csv`; per-genus B_{std} computed as the mean across all OTUs assigned to that genus.

Minimum sample filter: Genera detected in < 5 samples are excluded. Sensitivity checks at $n_{\min} = 50$ and $n_{\min} = 150$ produce identical results (analyses S4 and S5, Table S2 of the main text), confirming that most genera in the dataset exceed these thresholds.

3.3 EMP niche breadth (secondary, exploratory)

Data source: Earth Microbiome Project 16S data (`refdata.emp_16s`). EMPO (Earth Microbiome Project Ontology) level-3 habitat categories used as the niche axis; level-2 used as fallback when < 5 distinct EMPO-3 categories were present for a genus. Levins B_{std} computed from the genus $\times$ EMPO-category co-occurrence matrix. Minimum: ≥ 5 EMP samples per genus. $n = 539$ genera after merge with primary KO density data.

Result: $\lambda = 0.055$ (little phylogenetic signal in EMP niche breadth), $\beta = -0.019$, SE $= 0.0115$, $p = 0.099$ (non-significant, directionally consistent; see Fig. S13).

3.4 Soil-specialist definition

For the soil-restricted sensitivity analysis (analysis S7 of the main text), genera were classified as soil specialists via MicrobeAtlas `Env_Level_1` using ecosystem categories: *soil*, *agricultural*, *farm*, *field*, *paddy*, *peatland*, *desert*, *shrub*. Procedure: (1) identify the dominant `Env_Level_1` category per OTU (row with maximum `n_samples_detected`); (2) flag OTU as `is_soil` if dominant environment $\in$ SOIL_ENVS; (3) join OTUs to genera; (4) compute frac_soil $= n_{\text{soil OTUs}}/n_{\text{OTUs}}$ per genus; (5) classify genus as soil specialist if frac_soil > 0.5. Soil-restricted dataset: $n = 162$ genera.

SM4. Phylogenetic tree

Source: GTDB (Genome Taxonomy Database) Release 214v1 [7]. The full genus-level phylogeny was downloaded in Newick format and pruned to retain only genera present in the PGLS input dataset using `dendropy.Tree.prune_taxa_with_labels()`.

- Bacterial tree: `data/gtdb_bac_genus_pruned.tree` (2,283 taxa before pruning; 1,574 retained for the primary analysis).
- Archaeal tree: `data/gtdb_arc_genus_pruned.tree` (pruned to 95 genera for the archaeal replication; 41 for the archaeal sensitivity analysis at $n_{\min} = 5$ genera per taxon).

Tip labels use lowercase genus names with spaces replaced by underscores, matching the `genus_lower` column in all data files. All branch lengths are in units of substitutions per site (molecular phylogram; root-to-tip distances range from 0.886 to 2.994 substitutions per site, confirming the tree is **not ultrametric**).

Note on ultrametric assumption. PGLS under Brownian motion theoretically assumes an ultrametric (time-calibrated) phylogeny. The GTDB bacterial tree is a molecular phylogram, not a time-scaled tree, and this is a limitation acknowledged here. Two features of our implementation mitigate the impact: (1) Pagel's λ is estimated by maximum likelihood for each model, which scales off-diagonal covariances and partially corrects for the non-ultrametric structure [14]; (2) sensitivity analyses confirm that the primary result holds at $\lambda = 0$ (OLS; $\beta = -0.032$, $p \approx 0$) and $\lambda = 1$ (full Brownian motion; $\beta = -0.018$, $p = 3.7 \times 10^{-6}$), spanning the range from "tree structure ignored" to "full BM". The Grafen (1989) transformation to an arbitrary ultrametric tree was not applied; this is a future robustness direction.

The circular cladogram of the bacterial tree coloured by KO density and niche breadth is shown in Fig. S1.

SM5. Phylogenetic generalised least squares

5.1 Implementation

All PGLS models were fitted using `scripts/pgls_utils.py`, a custom implementation using `dendropy` 5.0.8 for tree operations and `scipy` 1.17.1 [8] / `numpy` 2.4.4 for optimisation and linear algebra. All analyses were run in Python 3.13.9 (conda-forge, GCC 14.3.0). CatBoost (version 1.2) was used for the exploratory LOPO analysis (SM 11). No R packages were used in the primary confirmatory analyses.

Variance–covariance matrix. Built from the pruned GTDB tree as $V[i, j]$ = shared branch length from root to MRCA(i, j) (= root-to-tip path length for $i = j$).

Pagel's λ. Off-diagonal elements of V are scaled by $\lambda \in (0, 1)$:

$$V_\lambda[i, j] = \lambda \cdot V[i, j] \quad (i \neq j); \qquad V_\lambda[i, i] = V[i, i] \tag{S4}$$

$\lambda = 0$: OLS (no phylogenetic signal); $\lambda = 1$: Brownian motion (full phylogenetic signal). λ is estimated by maximising the profile log-likelihood:

$$\log L = -\tfrac{1}{2}\left[n \log(2\pi) + n \log(\sigma^2) + \log |V_\lambda| + (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^\top V_\lambda^{-1} (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})/\sigma^2\right] \tag{S5}$$

Optimisation via `scipy.optimize.minimize_scalar()` with bounds (10^{-4}, $1-10^{-4}$) and `method='bounded'`.

GLS fit. After λ optimisation, GLS coefficients $\hat{\boldsymbol{\beta}}$ and standard errors are estimated:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top V^{-1}\mathbf{X})^{-1}\mathbf{X}^\top V^{-1}\mathbf{y} \tag{S6}$$

$$\sigma^2 = \frac{(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}})^\top V_\lambda^{-1}(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}})}{n - p} \tag{S7}$$

$$\mathrm{SE}(\hat{\beta}_j) = \sqrt{\left[(\mathbf{X}^\top V^{-1}\mathbf{X})^{-1}\right]_{jj} \cdot \sigma^2} \tag{S8}$$

where p is the number of predictors including the intercept.

Model statistics: $t_j = \hat{\beta}_j/\mathrm{SE}(\hat{\beta}_j)$ with df $= n - p$; AIC $= -2\log L + 2(p+1)$; ΔAIC $=$ AIC(model) $-$ AIC(null); $r^2 = 1 - \mathrm{Var(residuals)}/\mathrm{Var(response)}$.

5.2 Primary result

Table S3. Primary PGLS result (bacteria, $n = 1{,}574$ genera; MicrobeAtlas niche breadth vs Tier 1+2 KO density).

Parameter	Value
n genera (bacteria)	1,574
Pagel's λ	0.757
β (`ko_per_mb_primary`, z-scored)	-0.0207
SE	0.00368
t-statistic	-5.629
p (raw)	2.14×10^{-8}
p (FDR, joint with P2 and P3)	4.28×10^{-8}
r^2	0.046
ΔAIC vs null	-29.41
Archaeal replication ($n = 95$): β	-0.0137 ($p = 0.119$, FDR $= 0.178$)
Geochemical replication ($n = 482$): β	-0.00171 ($p = 0.755$)

SM6. Multiple testing correction

All corrections use the Benjamini–Hochberg (BH) procedure.

Primary three-way replication (§2.1 of main text). P-values from the three primary confirmatory tests (bacteria, archaea, geochemical) corrected jointly ($n = 3$). FDR: bacteria $= 4.28 \times 10^{-8}$; archaea $= 0.178$; geochemical $= 0.755$.

Evidence-tier comparisons (§2.2 of main text). Two tier comparisons (`all_non_ambiguous` and `bacmet_only`) corrected together ($n = 2$).

Functional category comparisons (§3 of main text). Five functional category PGLS models corrected together ($n = 5$).

Metal-specific analyses (§4 of main text). Nine metal-specific PGLS models corrected together ($n = 9$); all nine pass FDR < 0.05.

Clade-stratified analyses. Four phylum-stratified PGLS models corrected together ($n = 4$).

No multiple testing correction was applied to pre-specified confounder checks (§5.2 of the main text) or to the eight sensitivity analyses (§5.3), as these are robustness tests of a single primary result rather than independent hypotheses. This design choice is pre-registered: confounder checks each ask "does the primary result survive after accounting for variable X?"—they test whether β_{primary} is robust, not whether each confounder is independently associated with niche breadth. Applying BH-FDR across confounder checks would be statistically incorrect because the checks share the same null hypothesis (H_0: primary $\beta = 0$) and the confounder itself is not the quantity of interest. The split-magnitude permutation test (Section 7 of the main text) is treated as an independent hypothesis test and is therefore the sole focus of the confirmatory analysis family.

SM7. Confounder analyses

For each potential confounder, we refitted the PGLS model adding the confounder as a second predictor alongside `predictor_z` and recorded the change in the primary β:

$$\Delta\beta\,(\%) = \frac{|\hat{\beta}_{\text{with confounder}} - \hat{\beta}_{\text{baseline}}|}{|\hat{\beta}_{\text{baseline}}|} \times 100 \tag{S9}$$

Decision rule: $\Delta\beta > 50\%$ AND $p > 0.05$ for the density predictor $\Rightarrow$ confounder explains the effect.

Table S4. Confounder analysis results.

Confounder	Operationalisation	$\Delta\beta$ (%)	Verdict
Genome size	Mean genome size in Mb (z-scored)	46.7	ROBUST — below threshold; $p = 0.006$
GC content	Mean genomic GC% (z-scored)	23.7	ROBUST
Isolation source	Categorical dummy (culture environment)	14.5	ROBUST
Mean latitude	Mean absolute latitude per genus	51.8	ATTENUATED — β becomes *more* negative (-0.031), inconsistent with confounding; latitude amplifies rather than explains the signal
Dominant biome	Majority `Env_Level_1` (MicrobeAtlas)	5.8	ROBUST

No confounder meets the pre-specified decision rule ($\Delta\beta > 50\%$ AND non-significant). Genome size partially attenuates β by 46.7% but the predictor remains significant ($p = 0.006$), suggesting that genome size is a partial mediator rather than a confounder.

SM8. Sensitivity analyses

Eight pre-specified sensitivity analyses tested robustness to modelling assumptions. All used the same response variable (`mean_levins_B_std`) and the same primary KO density predictor, with the modification noted.

Table S5. Sensitivity analysis results. Six of eight analyses are directionally consistent with the primary result and statistically significant.

ID	Modification	β	p	Notes
S1	λ fixed = 0 (OLS)	-0.0319	≈ 0	Stronger effect under OLS
S2	λ fixed = 1 (Brownian)	-0.0182	3.66×10^{-6}	Consistent
S3	Archaea, $n_{\min} = 5$ genera	-0.0183	0.084	$n = 41$; directionally consistent, underpowered
S4	$n_{\min} = 50$ MicrobeAtlas samples	-0.0207	2.14×10^{-8}	Identical to primary
S5	$n_{\min} = 150$ MicrobeAtlas samples	-0.0207	2.14×10^{-8}	Identical to primary
S6	Raw (unstandardised) Levins B	-0.286	1.11×10^{-11}	Stronger signal; same direction
S7	Australia-only subset	-0.00171	0.755	Null; same result as geochemical replication
S8	Northern hemisphere only	-0.0302	3.24×10^{-6}	Stronger than primary; consistent

S4 and S5 produce identical results because all 1,574 genera in the primary dataset have MicrobeAtlas detection counts well above 150, making both thresholds redundant. S7 is numerically equivalent to the geochemical replication (same $n = 482$ Australian subset; Australia is represented in the global dataset but the narrowed geographic range removes the global diversity signal).

SM9. Pre-registration and analysis governance

9.1 Pre-registration

All confirmatory analyses (primary PGLS, archaeal replication, geochemical replication, evidence-tier comparisons, functional category comparisons, metal-specific comparisons, and all eight sensitivity analyses) were pre-specified in `RESEARCH_PLAN.md` prior to data analysis. This document recorded the hypothesis direction ($\beta < 0$), the response variable (`mean_levins_B_std`), the predictor (`ko_per_mb_primary`), the tree source (GTDB r214v1), the λ estimation method (ML), the multiple testing procedure (BH), and the decision rule for confounders ($\Delta\beta > 50\%$ AND $p > 0.05$).

Exploratory analyses (EMP niche breadth validation, BacDive geographic breadth, ENIGMA site-level replication, latitude mechanism tests, inverse PGLS, RDA variance partitioning, and community-weighted mean prediction) are labelled as such throughout the manuscript and were not pre-specified.

9.2 Analysis governance

Every reported statistic traces to a row in `INTERPRETATION_TABLE.md`, which records the notebook cell ID, output CSV file, and direction prediction for each confirmatory test. Statistical values were entered from saved CSV files in `data/`, not from notebook-cell display output. Exploratory analyses are tracked in a separate section of the same table, labelled "(Exploratory)".

SM10. Latitude mechanism tests

The latitude amplification result (adding mean absolute latitude as a covariate increases $|\beta|$ from -0.021 to -0.031, §5.2 of main text) motivated a set of nine mechanistic tests examining whether a specific environmental variable could explain the latitudinal gradient in niche breadth. These are exploratory analyses.

Models A–I each added a single environmental predictor alongside latitude to the primary PGLS, testing whether the latitude effect was absorbed by:

- **Model A (bedrock Cr composite)**: GeoROC lithospheric metal-index principal component (driven by Cr: BH $p = 6.7 \times 10^{-9}$, and Co: BH $p = 0.0084$).
- **Model B (temperature range)**: CHELSA annual temperature range (°C).
- **Model C–H**: Ore-deposit proximity; bioavailable metal mobility index; soil metal hazard (CSU classification); soil temperature; lithology class; REE deposit proximity.
- **Model I (soil organic matter)**: SOM (%) as a redox proxy. **Key result:** SOM is an independent negative predictor ($\beta = -0.016$, $p = 6.8 \times 10^{-5}$); absorbs the pH association (pH becomes non-significant when SOM is included), consistent with organic-matter-driven metal speciation and bioavailability.
- **Redox controls**: Soil moisture and SOM do not attenuate the Cr or Co bedrock signals ($p = 0.44$ and 0.50 for attenuation tests), arguing against a waterlogging / soil-reduction mechanism for the serpentine Cr effect.

All 28 extended models are in `data/latitude_mechanism_results.csv`.

SM11. Exploratory CatBoost LOPO analysis

These analyses are exploratory and carry lower inferential weight than the pre-registered confirmatory analyses H1–H6.

11.1 Design and software

CatBoost gradient-boosted trees (v1.2.10) [9] were applied in a Leave-One-Phylum-Out (LOPO) cross-validation design. LOPO held out all genera from one phylum at a time across 11 phyla (Proteobacteria 660, Firmicutes 330, Actinobacteria 203, Bacteroidetes 180, Cyanobacteria 27, Planctomycetes 17, Chloroflexi 15, Verrucomicrobia 14, Spirochaetes 11, Synergistetes 11, Acidobacteria 10 genera), fitting on the remaining genera and evaluating predictions on held-out genera using Spearman ρ. Model hyperparameters: `iterations=200, depth=4, learning_rate=0.05, l2_leaf_reg=3.0, thread_count=1, random_seed=42`. `thread_count=1` is required on the 128-core analysis server to prevent CatBoost from consuming all cores via implicit parallelism; the environment variable `OMP_NUM_THREADS=1` was set at launch.

11.2 Feature construction

KO presence records from `data/nb25_ko_presence_matrix.parquet` (118 Tier 1+2 KOs with $\geq$20 genera each) were pivoted to a genus $\times$ KO matrix, converted to per-Mb density (count of genomes with KO / mean genome Mb), and z-scored across genera. Genome size (`genome_size_mb_z`) was included as a covariate in all models. Functional subset z-scores (12 predictors) were used in the subset-level models; KO-to-subset assignments are from `data/s8_ko_metadata.csv`.

11.3 Environmental responses

Fifteen environmental z-scores were used as response variables: six CSU mobile metal fractions ($\mathrm{PF1}_{\mathrm{As,Cd,Cr,Cu,Hg,Pb}}$) computed from the global mobility grid [12]; six GeoROC bedrock metals (log Co, Cr, Cu, Ni, Pb, Zn) from `arkinlab_microbeatlas.enriched_metadata`; and Soil pH, ERA5 temperature, and Env PC1 (first PC of the eight metal+pH+temperature variables).

11.4 Model configurations

1. **All-KO LOPO** (Step 1a): 118 KO densities + genome size; $15 \times 11 = 165$ fits. Metric: Spearman ρ on held-out genera.
2. **Full-data SHAP** [10] (Step 1b): CatBoost fit on all genera per response; SHAP via `get_feature_importance(type='ShapValues')`.
3. **Single-KO LOPO** (Step 1c): two features (KO density + genome size) per KO $\times$ response $\times$ fold; $\approx$19,470 fits total; faster hyperparameters (`iterations=100`).
4. **Within-subset LOPO + SHAP** (Step 1d): four functional subsets (Resistance, Transport, Sensing, Cofactor) $\times$ 15 responses $\times$ 11 folds.
5. **Functional-subset LOPO + SHAP heatmap** (Steps 2a/2b): 12 functional-subset z-scores as features.
6. **Abundance-weighted comparison** (Step 3): sample weights $\log_{10}(n_{\mathrm{samples}} + 1)$, normalised to mean = 1.

7. **Reverse classifier** (Step 4): `CatBoostClassifier` predicts KO presence (binary) from the 15 environmental z-scores; LOPO AUC.

11.5 KO prioritisation composite score

For each of 118 KOs, four sub-scores were computed and then min–max normalised to $[0, 1]$ across all 118 KOs:

1. s_1: best $|\overline{\rho}|$ across all 15 responses (peak predictive power).
2. s_2: mean $|\overline{\rho}|$ across all 15 responses (breadth).
3. s_3: count of responses with $|\overline{\rho}| \geq 0.10$.
4. s_4: $-\log_{10}(\text{BH-FDR } q_{\min})$ from the per-KO PGLS on the three responses with available data (GEOROC Co, GEOROC Pb, Soil pH); zero if no PGLS p is available.

Composite score $= (s_1 + s_2 + s_3 + s_4)/4$. KEGG pathway annotations for the top 25 KOs were assigned from KEGG [11] database entries; the four module groupings (Porphyrin/Cobalamin, Cu/Ag efflux cus, Metal transport, Sensing/stress/other) reflect documented pathway membership. Five KOs were designated as validation priorities: K01772 (*hemH*), K09883 (*cobT*), K07796 (*cusC*), K19575 (*bmrR*), K09823 (*zur*).

11.6 Output files

File	Contents
`data/catboost_lopo_results.csv`	Per-response avg ρ; all-KO and subset models
`data/catboost_single_ko_ranking.csv`	Single-KO ρ per KO $\times$ response (1,770 rows)
`data/catboost_within_subset_shap.csv`	Per-KO SHAP within each subset
`data/catboost_shap_heatmap.csv`	Functional-subset SHAP matrix
`data/catboost_split_validation.csv`	Resistance/cofactor SHAP per response
`data/ko_prioritization_scores.csv`	Composite scores for 118 KOs
`data/kegg_pathway_overlay.csv`	Top-25 KO $\times$ 15-response ρ matrix

SM12. Social niche breadth benchmarking

SNB_{SES} was computed for all 3,389 genera detected in ≥ 10 MicrobeAtlas samples (of 3,433 genera total; 462,716 samples across the full MicrobeAtlas database). For each genus i with N_i occurrences, the raw SNB partner count is the number of other genera co-occurring in at least one shared sample. Because $S = 462{,}716$ makes the standard 999-permutation null model computationally infeasible, we used the exact analytical null: the probability that genus j is present in a random draw of N_i samples from S is $p_{ij} = 1 - (1 - N_j/S)^{N_i}$, giving null mean $\mu_i = \sum_{j \neq i} p_{ij}$ and null variance $\sigma_i^2 = \sum_{j \neq i} p_{ij}(1 - p_{ij})$ (Bernoulli indicator sum, exact as $S \to \infty$). SES $= (\text{raw} - \mu_i)/\sigma_i$. Per-genus metrics and PGLS inputs are saved in `data/snb_von_meijenfeldt_replication.csv`.

Spearman correlations between SNB_{SES} and our primary co-occurrence and niche-breadth metrics are given in Table S6. PGLS regressions of SNB_{SES} against metal-gene density (soil stratum, $n = 838$ genera, $\lambda = 0.81$) gave: Model 1, total KO density: $\beta = -0.35$, $p = 0.49$; Model 2, resistance vs. cofactor decomposition: $\beta_{\text{resist}} = +0.67$, $p = 0.22$; $\beta_{\text{cofactor}} = +0.68$, $p = 0.27$. Neither model reached significance, consistent with the interpretation that SNB_{SES} at this global scale reflects biogeographic range rather than local metal-gene niche structure.

Table S6. Spearman correlations between SNB_{SES} and alternative co-occurrence and niche-breadth metrics. "Sig. positive partners" = per-genus count of significantly co-occurring partners under the hypergeometric null (FDR < 5%, upper tail); Levins' B_{std} = standardised niche breadth from MicrobeAtlas biome profiles.

Comparison	ρ	p	n
SNB_{SES} vs. phi-coefficient degree	-0.207	4.9×10^{-34}	3,389
SNB_{SES} vs. sig. positive partners	$+0.340$	2.0×10^{-92}	3,389
SNB_{SES} vs. Levins' B_{std}	-0.184	1.7×10^{-5}	539

SM13. Two-scale phylogenetic signal: method detail and validity assessment

13.1 Method overview and novelty

The two-scale framework combines genus-level Pagel's λ (applied to continuous genus presence fractions) with genome-level Fritz & Purvis D (applied to binary genome presence/absence) on the same set of 275 curated metal KOs. This specific combination does not appear in the literature we searched; the closest conceptual analog pairs two distinct evolutionary signals (synteny index and sequence-mutability measure) to detect HGT between closely related taxa [16]. The general logic—that different methods detect transfers at different evolutionary timescales, with composition/signal-based methods sensitive to recent transfers and phylogenetic/topology-based methods to ancient fixed transfers—is established across the HGT-detection literature. λ here captures deep, genus-level phylogenetic clustering (trait conserved across millions of years of vertical transmission); D captures genome-level dispersal (recent or episodic horizontal acquisition detectable as phylogenetic randomness within and across genera).

13.2 D computation: single pooled genome tree

Fritz & Purvis D was estimated once per KO on a single pooled 18,961-tip GTDB genome tree (all bacterial genomes in the dataset), not per-genus. For each KO, binary presence/absence was coded across all 18,961 genome tips; the observed ancestral-state-change statistic was calibrated against Brownian-motion ($D = 0$ expectation) and random-permutation ($D = 1$ expectation) null distributions, each with $n = 1{,}000$ simulations. This produces a single D value per KO, interpretable as a spectrum from phylogenetically clustered ($D \approx 0$, Brownian-motion-like) to phylogenetically dispersed ($D \approx 1$, random).

The choice of a pooled tree over per-genus trees is a deliberate design decision: it tests for genome-level randomness across the full phylogenetic depth of the dataset, capturing both deep and shallow dispersal, rather than per-genus dispersal which would require a separate tree and sufficient sample size for each genus. The trade-off is that small genera (few genomes) may have limited power to contribute meaningful signal to the genome-scale D estimate; because D is computed on the pooled tree, this is less of a per-genus issue than for λ.

13.3 Validity concern 1: Pagel's λ on bounded presence-fraction data

Pagel's λ was designed for continuous traits evolving under Brownian motion (BM)—an unconstrained Gaussian process. Genus presence fractions are bounded proportions in $[0, 1]$, violating the Gaussian/BM assumption in two specific ways:

1. **Boundary compression.** Near-core genes (fraction $\to$ 1) and rare genes (fraction $\to$ 0) cannot vary freely; λ estimation assumes residuals can take any real value, which is false at the boundaries. Many genes in a curated 275-KO metal-gene set will fall near one boundary or the other (core cofactor genes near 1.0; rare resistance genes near 0.0), making this a particularly relevant concern.

2. **Unequal measurement precision.** A genus with $n = 2$ genomes can only yield presence fractions of 0, 0.5, or 1 — completely discrete — while a genus with $n = 200$ genomes provides a continuous fraction with real precision. Standard PGLS/λ estimation does not downweight low-precision tips unless genome-count-weighted measurement error is explicitly modelled. In the primary dataset, 308 of 1,574 genera have exactly two genomes ($n_{\text{genomes}} = 2$ minimum),

potentially injecting discretisation noise at the tips.

Robustness checks (performed). We applied two checks to assess whether the bounded-proportion structure drives our λ estimates.

Check A: Arcsine-square-root transformation (Python, $n = 88$ KOs). Genus presence fractions were arcsine-square-root-transformed before λ estimation. Re-estimating λ for all 88 Tier 1–2 KOs with sufficient genus representation yielded Spearman $r = 0.613$ ($p < 0.001$) between original and transformed estimates (mean $|\Delta\lambda| = 0.195$; median 0.179). Nine KOs (10.2%) crossed the $\lambda = 0.3$ weak/strong classification threshold, and *all nine* changed in the weak-to-strong direction, indicating that the original PGLS *underestimates* phylogenetic signal for sparse genes rather than overestimating it. The shifts are concentrated in transport and resistance genes with very low original λ (e.g., *mntC*: $0.000 \to 0.525$; *troA*: $0.143 \to 0.683$; *shp*: $0.000 \to 0.552$). None of the 13 double-signal KOs (low λ and high D) changed classification under this transformation.

Check B: Phylogenetic lambda model on transformed fractions (`phylolm` in R, $n = 33$ KOs). We fitted `phylolm` with `model="lambda"` to arcsine-square-root-transformed fractions for 33 KOs for which all required data were available. Spearman $\rho = 0.174$ between original PGLS λ and `phylolm` λ (mean $|\Delta\lambda| = 0.281$; 10/33, 30.3% classification changes), with all changes again in the weak-to-strong direction. The lower concordance relative to Check A likely reflects the smaller matched sample ($n = 33$ KOs) as much as methodological difference; the directional consistency confirms the conservative bias of the original estimates.

Conclusion. Both transformation approaches reveal that original λ estimates *moderately underestimate* phylogenetic signal for sparse genes. Because all classification shifts move weak-to-strong, our identification of cofactor KOs as high-λ and resistance KOs as low-λ is conservative: the 13 double-signal (low-λ, high-D) resistance KOs have at least as much phylogenetic clustering as stated. Beta-regression alternatives (e.g., `betareg` in R) were not available in our compute environment and remain a priority for future independent replication.

13.4 Validity concern 2: sparse-genus noise and the orthogonality result

The Spearman correlation between D and λ across 275 KOs is $\rho = -0.041$ ($p = 0.49$), reported in the main text as consistent with the two metrics capturing distinct evolutionary processes. However, a near-zero correlation has two equally plausible interpretations:

- **True independence:** D and λ capture biologically distinct signals (dispersal timescale) and are genuinely orthogonal.
- **Noise dominance:** One or both metrics are dominated by estimation variance (particularly for λ estimated from sparse-genome genera or small-KO-presence KOs), and the near-zero ρ reflects the absence of a detectable signal in either direction rather than biological independence.

Distinguishing these interpretations requires a threshold stability check: recompute ρ after successively removing genera below minimum-genome-count thresholds ($n_{\min} = 2, 5, 10, 20$); if ρ remains near zero, sparse-genome genera are not masking a real correlation.

Robustness check (performed). We built a complete genus $\times$ KO presence matrix for 240 of the 275 metal KOs from 16 million genome records (MGnify MAGs and SPIRE prokaryotic genomes combined) and recomputed $\rho(\lambda, D)$ at each threshold.

$n_{\min}$	Genera retained	KOs with data	ρ	p
2	8,256	238	−0.068	0.295
5	3,799	237	−0.020	0.765
10	2,166	236	−0.097	0.138
20	1,204	232	+0.100	0.129

The range across thresholds is $\Delta\rho = 0.197$ and no threshold yields a significant correlation (all $p > 0.12$). Importantly, ρ does not trend monotonically toward a detectable signal as sparse genera are removed; the values oscillate near zero throughout. This confirms that the near-zero ρ is not an artefact of sparse-genome genera dominating either metric: when low-genome-count genera are progressively excluded, no latent correlation emerges. The two-scale framework's orthogonality result is therefore robust, supporting the interpretation that λ and D capture biologically distinct evolutionary signals at different timescales.

13.5 Comparison with Panstripe

Panstripe [17] is a generalised-linear-regression-based tool designed specifically for phylogenetically controlled analysis of gene gain and loss rates in prokaryotic pangenomes. It is robust to population structure, sampling bias, and errors in predicted presence/absence of genes—all concerns that apply to the D statistic as applied here. Panstripe was not applied in this analysis primarily because it requires a per-gene presence/absence matrix with a shared pangenome phylogeny, whereas our D estimation used a pooled genome tree from GTDB. Applying Panstripe to the 140 primary metal KOs and comparing gene-gain/loss rates between cofactor biosynthesis and resistance subcategories would provide a more rigorous, purpose-built alternative to the D-based HGT screen and is an explicit future priority.

Supplementary Tables

Table S2. Primary 140-KO metal-gene list with evidence tier, functional category, metal targets, and source databases. Tier: T1 = Tier 1 (multi-source validated, $n = 32$); T2 = Tier 2 (KEGG module + one additional source, $n = 108$). Category abbreviations: Transport = Transport/Homeostasis; Resistance = Resistance/Detoxification; Sensing = Sensing/Regulation; Cofactor = Cofactor Biosynthesis; Metabolism = Metal-dependent Metabolism. Source: KEGG = KEGG module annotation; BacMet2 = BacMet v2.0 database; Fitness = FitnessBrowser transposon screen.

KO	Gene	Tier	Category	Metals	Source
K01534	zntA	T1	Transport	Zn, Cd, Cu, Ni, Co, Hg, Pb	BacMet2, Fitness
K02006	cbiO	T2	Transport	Co, Ni	—
K02007	cbiM	T2	Transport	Co, Ni	—
K02008	cbiQ	T2	Transport	Co, Ni	—
K02009	cbiN	T2	Transport	Co, Ni	—
K02010	afuC	T1	Transport	Fe, Co, Zn	BacMet2, Fitness
K02012	afuA	T1	Transport	Fe, Co, Zn	BacMet2, Fitness
K02013	ABC.FEV.A	T2	Transport	Fe	—
K02015	ABC.FEV.P	T2	Transport	Fe	—
K02016	ABC.FEV.S	T1	Transport	Fe	BacMet2
K02068	STAR1	T1	Transport	Fe, Ni	BacMet2, Fitness
K02069	STAR2	T1	Transport	Fe, Ni	BacMet2, Fitness
K02074	ABC.ZM.A	T2	Transport	Mn, Zn	—
K02075	ABC.ZM.P	T2	Transport	Mn, Zn	—
K02077	ABC.ZM.S	T2	Transport	Mn, Zn	—
K02190	cbiK	T2	Transport	Co	KEGG
K02230	cobN	T2	Transport	Co	KEGG
K03523	bioY	T2	Transport	—	—
K03795	cbiX	T2	Transport	Co	KEGG
K05895	cobK-cbiJ	T2	Transport	Co	KEGG
K08197	ARN	T2	Transport	Fe	—
K08225	entS	T2	Transport	Fe	—
K08970	rcnA	T1	Transport	Co, Ni	BacMet2, Fitness
K09815	znuA	T1	Transport	Zn	BacMet2
K09816	znuB	T1	Transport	Zn	BacMet2
K09817	znuC	T1	Transport	Zn	BacMet2
K09818	ABC.MN.S	T2	Transport	Mn, Fe	—
K09819	ABC.MN.P	T2	Transport	Mn, Fe	—
K09820	ABC.MN.A	T2	Transport	Mn, Fe	—
K09882	cobS	T2	Transport	Co	KEGG
K09883	cobT	T2	Transport	Co	KEGG
K10094	cbiK	T2	Transport	Ni	—
K10830	psaB	T2	Transport	Mn, Zn	—
K10834	hmuV	T2	Transport	Fe	—
K11601	mntC	T2	Transport	Mn	—
K11602	mntB	T2	Transport	Mn	—

Continued on next page.

Table S2 continued.

KO	Gene	Tier	Category	Metals	Source
K11603	mntA	T2	Transport	Mn	—
K11604	sitA	T1	Transport	Mn, Fe	BacMet2
K11605	sitC	T1	Transport	Mn, Fe	BacMet2
K11606	sitD	T1	Transport	Mn, Fe	BacMet2
K11607	sitB	T1	Transport	Mn, Fe	BacMet2
K11704	mtsA	T2	Transport	Mn, Cu, Fe, Zn	—
K11705	mtsC	T2	Transport	Mn, Cu, Fe, Zn	—
K11706	mtsB	T2	Transport	Mn, Cu, Fe, Zn	—
K11707	troA	T1	Transport	Mn, Fe, Zn	BacMet2
K11708	troC	T1	Transport	Mn, Fe, Zn	BacMet2
K11709	troD	T1	Transport	Mn, Fe, Zn	BacMet2
K11710	troB	T1	Transport	Mn, Fe, Zn	BacMet2
K12346	SMF	T2	Transport	Fe	—
K14193	isdA	T2	Transport	Fe	—
K14709	SLC39A1_2_3	T2	Transport	Zn	—
K14710	SLC39A4	T2	Transport	Zn	—
K14711	SLC39A5	T2	Transport	Zn	—
K14712	SLC39A6	T2	Transport	Zn	—
K14713	SLC39A7	T2	Transport	Zn	—
K14714	SLC39A8	T2	Transport	Zn	—
K14715	SLC39A9	T2	Transport	Zn	—
K14716	SLC39A10	T2	Transport	Zn	—
K14717	SLC39A11	T2	Transport	Zn	—
K14718	SLC39A12	T2	Transport	Zn	—
K14719	SLC39A13	T2	Transport	Zn	—
K14720	SLC39A14	T2	Transport	Zn	—
K16302	CNNM	T2	Transport	—	—
K16783	bioN	T2	Transport	—	—
K16784	bioM	T2	Transport	—	—
K16915	cbiL	T2	Transport	Ni	—
K19971	psaA	T2	Transport	Mn, Zn	—
K19972	psaC	T2	Transport	Mn, Zn	—
K19973	mntA	T2	Transport	Mn	—
K19975	mntC	T2	Transport	Mn	—
K19976	mntB	T2	Transport	Mn	—
K20170	nicA1	T2	Transport	—	KEGG
K20982	LOVC	T2	Transport	—	KEGG
K22011	cfbA	T2	Transport	Co	KEGG
K23185	fepB	T2	Transport	—	—
K23186	fepD	T2	Transport	Fe	—
K23187	fepG	T2	Transport	Fe	—
K23188	fepC	T2	Transport	Fe	—
K24175	MFSD5	T2	Transport	Mo	—
K25027	btuC	T2	Transport	Co	—

Continued on next page.

Table S2 continued.

KO	Gene	Tier	Category	Metals	Source
K25028	btuD	T2	Transport	Co	—
K25034	btuF	T2	Transport	Co	—
K25109	sirA	T2	Transport	Fe	—
K25110	sirB	T2	Transport	Fe	—
K25111	sirC	T2	Transport	Fe	—
K25113	htsA	T2	Transport	Fe	—
K25114	htsB	T2	Transport	Fe	—
K25115	htsC	T2	Transport	Fe	—
K25116	isdD	T2	Transport	Fe	—
K25117	isdB	T2	Transport	Fe	—
K25118	isdC	T2	Transport	Fe	—
K25119	shp	T2	Transport	Fe	—
K25130	fecE	T2	Transport	Fe	—
K25132	hmuT	T2	Transport	Fe	—
K25133	hmuU	T2	Transport	Fe	—
K25282	yclQ	T2	Transport	Fe	—
K25283	yclO	T2	Transport	Fe	—
K25284	yclN	T2	Transport	Fe	—
K25285	yclP	T2	Transport	Fe	—
K25286	fagD	T2	Transport	Fe	—
K25287	desE	T2	Transport	Fe	—
K25288	fhuB	T2	Transport	Fe	—
K25289	fhuG	T2	Transport	Fe	—
K25290	fhuA	T2	Transport	Fe	—
K25308	fhuD	T2	Transport	Fe	—
K28864	YSL-PS	T2	Transport	—	—
K03325	ACR3	T1	Resistance	As, Tl	BacMet2, Fitness
K03446	emrB	T2	Resistance	Cu, Tl	BacMet2, Fitness
K07665	cusR	T1	Resistance	Cu, Ni	BacMet2, Fitness
K07785	nrsD	T1	Resistance	Ni	Fitness
K07787	cusA	T1	Resistance	Cu, Ag	BacMet2, Fitness
K07798	cusB	T1	Resistance	Cu, Ag, Zn	BacMet2, Fitness
K08365	merR	T1	Resistance	Hg, Cu, Zn	BacMet2, Fitness
K15725	czcC	T1	Resistance	Ni, Zn, Co, Cd	BacMet2, Fitness
K15726	czcA	T1	Resistance	Co, Ni, Zn, Cd	BacMet2, Fitness
K15727	czcB	T1	Resistance	Co, Ni, Zn, Cd	BacMet2, Fitness
K16264	czcD	T1	Resistance	Cd, Co, Zn, Tl	BacMet2, Fitness
K17686	copA	T2	Resistance	Cu	BacMet2, Fitness
K19591	cueR	T1	Resistance	Cu, Co, Ni, Zn	BacMet2, Fitness
K19594	gesB	T1	Resistance	Cu, Au	BacMet2
K19595	gesA	T1	Resistance	Cu, Au	BacMet2
K00240	sdhB	T2	Sensing	Fe, S	KEGG
K00245	frdB	T2	Sensing	Fe, S	KEGG

Continued on next page.

Table S2 continued.

KO	Gene	Tier	Category	Metals	Source
K02483	—	T2	Sensing	Al, Co, Cu	BacMet2, Fitness
K07644	cusS	T1	Sensing	Cu	BacMet2, Fitness
K08941	pscB	T2	Sensing	Fe, S	KEGG
K13638	zntR	T2	Sensing	Zn, Cu	BacMet2, Fitness
K17219	sreA	T2	Sensing	Mo, S	KEGG
K17220	sreB	T2	Sensing	S	KEGG
K17221	sreC	T2	Sensing	S	KEGG
K18367	CoADR	T2	Sensing	S	KEGG
K02225	cobC1	T2	Cofactor	Co	KEGG
K03635	MOCS2B	T2	Cofactor	Mo	KEGG
K03638	moaB	T2	Cofactor	Mo	KEGG
K03750	moeA	T2	Cofactor	Mo	KEGG
K03831	mogA	T2	Cofactor	Mo	KEGG
K00380	cysJ	T2	Metabolism	S, Co, Cu	KEGG, Fitness
K02484	—	T2	—	Al, Cu	BacMet2, Fitness
K03673	dsbA	T2	—	Ag, Al, Bi, Zn	BacMet2, Fitness
K03981	dsbC	T2	—	Al, Cu	BacMet2, Fitness

Table S3. Double-signal HGT candidate KOs: genome-level Fritz & Purvis D > 0.2 **and** genus-level Pagel's $\lambda < 0.3$. D was computed on the 18,961-tip GTDB genome tree using binary presence/absence; λ was estimated from genus presence fractions on the GTDB genus-level phylogeny. Low D indicates phylogenetically dispersed binary presence at the genome scale; low λ indicates low genus-level signal in continuous presence fractions. KOs are sorted by D (descending).

KO	Gene	Subcategory	Tier	D	λ	Metal relevance
K07785	*nrsD*	Resistance/Detox.	T2	0.821	0.089	Ni resistance (NrsD repressor)
K19059	*merE*	Transport/Homeostasis	T2	0.728	0.102	Hg transport (mer operon)
K19057	*merD*	Resistance/Detox.	T2	0.701	0.165	Hg resistance (mer operon)
K19594	*gesB*	Resistance/Detox.	T2	0.597	0.156	Au/Te resistance
K08356	*aoxB*	Resistance/Detox.	T2	0.562	0.000	As(III) oxidase
K19595	*gesA*	Resistance/Detox.	T2	0.458	0.161	Au/Te resistance
K25119	*shp*	Transport/Homeostasis	T3	0.385	0.000	Fe transport
K03897	*iucD*	Unknown	T3	0.354	0.237	Fe siderophore biosynthesis
K19592	*golS*	Sensing/Regulation	T2	0.265	0.135	Au/Te sensing
K05908	*doxDA*	Metal-dep. Metabolism	T3	0.254	0.000	As oxidation (sox pathway)
K08170	*norB*	Resistance/Detox.	T3	0.239	0.033	Cu/NO detoxification
K14974	*nicC*	Transport/Homeostasis	T3	0.224	0.000	Ni transport
K15585	*nikB*	Transport/Homeostasis	T3	0.202	0.000	Ni transport

Category distribution of double-signal HGT candidates. Among the 13 KOs meeting the double-signal criterion (D > 0.2 and $\lambda < 0.3$): 6 are Resistance/Detoxification (*nrsD*, *merD*, *gesA*, *gesB*, *aoxB*, *norB*); 4 are Transport/Homeostasis (*merE*, *shp*, *nicC*, *nikB*); 1 is Sensing/Regulation (*golS*); 1 is Metal-dependent Metabolism (*doxDA*); 1 is Unknown (*iucD*). **Zero** Cofactor Biosynthesis KOs meet the double-signal criterion, consistent with cofactor biosynthesis genes being vertically inherited and constitutively coupled to genome size rather than acquired by horizontal gene transfer.

Table S4. Curation rationale for KOs assigned to thallium (Tl), aluminium (Al), and sulfur (S): elements with fewer dedicated resistance systems whose KOs were assigned based on documented biochemical function or fitness-screen evidence. Tier codes: T1 = direct biochemical evidence; T2 = strong biochemical inference; T2F = fitness-screen evidence; T3 = computational/BacMet annotation. n = 153 KOs (Al = 34, Tl = 45, S = 74).

KO	Gene	Metal	Subcategory	Tier	Biochemical rationale
K02483	—	Al	Sensing/Regulation	T2	Sensing/regulatory response to Al,Co,Cu
K02484	—	Al	Unknown	T2	Fitness screen: defect under Al,Cu stress
K03673	*dsbA*	Al	Unknown	T2	Fitness screen: defect under Ag,Al,Bi,Zn stress
K03981	*dsbC*	Al	Unknown	T2	Fitness screen: defect under Al,Cu stress
K00528	*fpr*	Al	Unknown	T2F	Fitness screen: defect under Al,Co,Tl,Zn stress
K01002	*mdoB*	Al	Unknown	T2F	Fitness screen: defect under Al,Bi,Fe,Tl stress
K01069	*gloB*	Al	Unknown	T2F	Fitness screen: defect under Al,Tl,Zn stress
K01073	*yigI*	Al	Unknown	T2F	Fitness screen: defect under Al,Co,Cu,Ni,Tl stress
K01104	*E3.1.3.48*	Al	Unknown	T2F	Fitness screen: defect under Al stress
K01256	*pepN*	Al	Unknown	T2F	Fitness screen: defect under Al,Ni stress
K01262	*pepP*	Al	Unknown	T2F	Fitness screen: defect under Al,Cu,Ni stress
K01414	*prlC*	Al	Unknown	T2F	Fitness screen: defect under Al,Cu,Ni stress
K01626	*E2.5.1.54*	Al	Unknown	T2F	Fitness screen: defect under Al stress
K01784	*galE*	Al	Resistance/Detox.	T2F	Resistance/detoxification; fitness defect under Al,Tl,Zn stress
K02027	*ABC.MS.S*	Al	Unknown	T2F	Fitness screen: defect under Al,Cu,Ni stress
K02028	*ABC.PA.A*	Al	Unknown	T2F	Fitness screen: defect under Al,Ni stress

continued on next page

Table S4 continued.

KO	Gene	Metal	Subcategory	Tier	Biochemical rationale
K02030	*ABC.PA.S*	Al	Unknown	T2F	Fitness screen: defect under Al,Ni stress
K02424	*tcyA*	Al	Transport/Homeostasis	T2F	Metal transport/homeostasis (Al,Cu,Ni)
K03296	*TC.HAE1*	Al	Unknown	T2F	Fitness screen: defect under Al,Cu,Ni stress
K03316	*nhaP*	Al	Unknown	T2F	Fitness screen: defect under Al,Tl stress
K03760	*eptA*	Al	Unknown	T2F	Fitness screen: defect under Al stress
K03781	*katE*	Al	Unknown	T2F	Fitness screen: defect under Al,Ni stress
K03814	*mtgA*	Al	Unknown	T2F	Fitness screen: defect under Al,Cr,Mn stress
K03885	*ndh*	Al	Unknown	T2F	Fitness screen: defect under Al,Tl stress
K05810	*LACC1*	Al	Unknown	T2F	Fitness screen: defect under Al,Tl stress
K07165	*fecR*	Al	Unknown	T2F	Fitness screen: defect under Al,Bi,Tl stress
K07277	*SAM50*	Al	Unknown	T2F	Fitness screen: defect under Al,Ni stress
K07278	*tamA*	Al	Unknown	T2F	Fitness screen: defect under Al,Ni stress
K07338	—	Al	Unknown	T2F	Fitness screen: defect under Al,Co,Ni stress
K07636	*phoR*	Al	Sensing/Regulation	T2F	Sensing/regulatory response to Al,Cu
K07666	*qseB*	Al	Sensing/Regulation	T2F	Sensing/regulatory response to Al,Cu
K09800	*tamB*	Al	Unknown	T2F	Fitness screen: defect under Al,Co,Ni stress
K19353	*eptC*	Al	Unknown	T2F	Fitness screen: defect under Al stress
K20534	*gtrB*	Al	Unknown	T2F	Fitness screen: defect under Al,Tl stress
K00240	*sdhB*	S	Sensing/Regulation	T2	Sensing/regulatory response to Fe,S
K00245	*frdB*	S	Sensing/Regulation	T2	Sensing/regulatory response to Fe,S
K00380	*cysJ*	S	Metal-dep. Metabolism	T2	Metal-dependent metabolic enzyme (S,Co,Cu)
K08941	*pscB*	S	Sensing/Regulation	T2	Sensing/regulatory response to Fe,S

continued on next page

Table S4 continued.

KO	Gene	Metal	Subcategory	Tier	Biochemical rationale
K17219	*sreA*	S	Sensing/Regulation	T2	Sensing/regulatory response to Mo,S
K17220	*sreB*	S	Sensing/Regulation	T2	Sensing/regulatory response to S
K17221	*sreC*	S	Sensing/Regulation	T2	Sensing/regulatory response to S
K18367	*CoADR*	S	Sensing/Regulation	T2	Sensing/regulatory response to S
K00024	*mdh*	S	Metal-dep. Metabolism	T3	Metal-dependent metabolic enzyme (S)
K00031	*IDH1*	S	Metal-dep. Metabolism	T3	Metal-dependent metabolic enzyme (S)
K00169	*porA*	S	Unknown	T3	Fitness screen: defect under S stress
K00170	*porB*	S	Unknown	T3	Fitness screen: defect under S stress
K00171	*porD*	S	Unknown	T3	Fitness screen: defect under S stress
K00172	*porC*	S	Unknown	T3	Fitness screen: defect under S stress
K00174	*korA*	S	Unknown	T3	Fitness screen: defect under S stress
K00175	*korB*	S	Unknown	T3	Fitness screen: defect under S stress
K00176	*korD*	S	Unknown	T3	Fitness screen: defect under S stress
K00177	*korC*	S	Unknown	T3	Fitness screen: defect under S stress
K00239	*sdhA*	S	Metal-dep. Metabolism	T3	Metal-dependent metabolic enzyme (S)
K00241	*sdhC*	S	Metal-dep. Metabolism	T3	Metal-dependent metabolic enzyme (S)
K00242	*sdhD*	S	Metal-dep. Metabolism	T3	Metal-dependent metabolic enzyme (S)
K00244	*frdA*	S	Metal-dep. Metabolism	T3	Metal-dependent metabolic enzyme (S)
K00246	*frdC*	S	Metal-dep. Metabolism	T3	Metal-dependent metabolic enzyme (S)
K00247	*frdD*	S	Metal-dep. Metabolism	T3	Metal-dependent metabolic enzyme (S)
K00381	*cysI*	S	Metal-dep. Metabolism	T3	Metal-dependent metabolic enzyme (S)
K00390	*cysH*	S	Unknown	T3	Fitness screen: defect under S stress

continued on next page

Table S4 continued.

KO	Gene	Metal	Subcategory	Tier	Biochemical rationale
K00392	*sir*	S	Metal-dep. Metabolism	T3	Metal-dependent metabolic enzyme (S)
K00860	*cysC*	S	Unknown	T3	Fitness screen: defect under S stress
K00955	*cysNC*	S	Unknown	T3	Fitness screen: defect under S stress
K00956	*cysN*	S	Unknown	T3	Fitness screen: defect under S stress
K00957	*cysD*	S	Unknown	T3	Fitness screen: defect under S stress
K00958	*sat*	S	Unknown	T3	Fitness screen: defect under S stress
K01006	*ppdK*	S	Unknown	T3	Fitness screen: defect under S stress
K01007	*pps*	S	Unknown	T3	Fitness screen: defect under S stress
K01595	*ppc*	S	Unknown	T3	Fitness screen: defect under S stress
K01676	*E4.2.1.2A*	S	Unknown	T3	Fitness screen: defect under S stress
K01677	*E4.2.1.2AA*	S	Unknown	T3	Fitness screen: defect under S stress
K01678	*E4.2.1.2AB*	S	Unknown	T3	Fitness screen: defect under S stress
K01679	*E4.2.1.2B*	S	Unknown	T3	Fitness screen: defect under S stress
K01681	*ACO*	S	Unknown	T3	Fitness screen: defect under S stress
K01682	*acnB*	S	Unknown	T3	Fitness screen: defect under S stress
K01902	*sucD*	S	Unknown	T3	Fitness screen: defect under S stress
K01903	*sucC*	S	Unknown	T3	Fitness screen: defect under S stress
K01958	*PC*	S	Unknown	T3	Fitness screen: defect under S stress
K01959	*pycA*	S	Unknown	T3	Fitness screen: defect under S stress
K01960	*pycB*	S	Unknown	T3	Fitness screen: defect under S stress
K02045	*cysA*	S	Unknown	T3	Fitness screen: defect under S stress
K02046	*cysU*	S	Transport/Homeostasis	T3	Metal transport/homeostasis (S)

continued on next page

Table S4 continued.

KO	Gene	Metal	Subcategory	Tier	Biochemical rationale
K02047	*cysW*	S	Transport/Homeostasis	T3	Metal transport/homeostasis (S)
K02048	*cysP*	S	Unknown	T3	Fitness screen: defect under S stress
K03737	*por*	S	Unknown	T3	Fitness screen: defect under S stress
K05908	*doxDA*	S	Metal-dep. Metabolism	T3	Metal-dependent metabolic enzyme (S)
K08940	*pscA*	S	Unknown	T3	Fitness screen: defect under S stress
K08942	*pscC*	S	Unknown	T3	Fitness screen: defect under S stress
K08943	*pscD*	S	Unknown	T3	Fitness screen: defect under S stress
K13811	*PAPSS*	S	Unknown	T3	Fitness screen: defect under S stress
K15230	*aclA*	S	Unknown	T3	Fitness screen: defect under S stress
K15231	*aclB*	S	Unknown	T3	Fitness screen: defect under S stress
K15232	*ccsA*	S	Unknown	T3	Fitness screen: defect under S stress
K15233	*ccsB*	S	Unknown	T3	Fitness screen: defect under S stress
K15234	*ccl*	S	Unknown	T3	Fitness screen: defect under S stress
K16936	*doxA*	S	Metal-dep. Metabolism	T3	Metal-dependent metabolic enzyme (S)
K16937	*doxD*	S	Metal-dep. Metabolism	T3	Metal-dependent metabolic enzyme (S)
K17222	*soxA*	S	Unknown	T3	Fitness screen: defect under S stress
K17223	*soxX*	S	Unknown	T3	Fitness screen: defect under S stress
K17224	*soxB*	S	Unknown	T3	Fitness screen: defect under S stress
K17225	*soxC*	S	Metal-dep. Metabolism	T3	Metal-dependent metabolic enzyme (S)
K17226	*soxY*	S	Sensing/Regulation	T3	Sensing/regulatory response to S
K17227	*soxZ*	S	Sensing/Regulation	T3	Sensing/regulatory response to S
K17229	*fccB*	S	Metal-dep. Metabolism	T3	Metal-dependent metabolic enzyme (S)

continued on next page

Table S4 continued.

KO	Gene	Metal	Subcategory	Tier	Biochemical rationale
K17230	*fccA*	S	Metal-dep. Metabolism	T3	Metal-dependent metabolic enzyme (S)
K19713	*tsdA*	S	Metal-dep. Metabolism	T3	Metal-dependent metabolic enzyme (S)
K22622	*soxD*	S	Unknown	T3	Fitness screen: defect under S stress
K27925	*tetH*	S	Unknown	T3	Fitness screen: defect under S stress
K03325	*ACR3*	Tl	Resistance/Detox.	T1	Resistance/detoxification; fitness defect under As,Tl stress
K16264	*czcD*	Tl	Resistance/Detox.	T1	Resistance/detoxification; fitness defect under Cd,Co,Zn,Tl stress
K03446	*emrB*	Tl	Resistance/Detox.	T2	Resistance/detoxification; fitness defect under Cu,Tl stress
K00059	*fabG*	Tl	Unknown	T2F	Fitness screen: defect under Bi,Ni,Tl stress
K00108	*betA*	Tl	Metal-dep. Metabolism	T2F	Metal-dependent metabolic enzyme (Tl)
K00130	*betB*	Tl	Metal-dep. Metabolism	T2F	Metal-dependent metabolic enzyme (Tl)
K00523	*ascD*	Tl	Unknown	T2F	Fitness screen: defect under Co,Cu,Tl stress
K00528	*fpr*	Tl	Unknown	T2F	Fitness screen: defect under Al,Co,Tl,Zn stress
K01002	*mdoB*	Tl	Unknown	T2F	Fitness screen: defect under Al,Bi,Fe,Tl stress
K01069	*gloB*	Tl	Unknown	T2F	Fitness screen: defect under Al,Tl,Zn stress
K01073	*yigI*	Tl	Unknown	T2F	Fitness screen: defect under Al,Co,Cu,Ni,Tl stress
K01278	*DPP4*	Tl	Unknown	T2F	Fitness screen: defect under Cu,Tl stress
K01782	*fadJ*	Tl	Resistance/Detox.	T2F	Resistance/detoxification; fitness defect under Co,Tl stress
K01784	*galE*	Tl	Resistance/Detox.	T2F	Resistance/detoxification; fitness defect under Al,Tl,Zn stress
K01814	*hisA*	Tl	Resistance/Detox.	T2F	Resistance/detoxification; fitness defect under Co,Tl stress
K01990	*ABC-2.A*	Tl	Unknown	T2F	Fitness screen: defect under Co,Fe,Ni,Tl stress
K02072	*metI*	Tl	Transport/Homeostasis	T2F	Metal transport/homeostasis (Cr,Cu,Tl)

continued on next page

Table S4 continued.

KO	Gene	Metal	Subcategory	Tier	Biochemical rationale
K02167	*betI*	Tl	Sensing/Regulation	T2F	Sensing/regulatory response to Tl
K02493	*hemK*	Tl	Unknown	T2F	Fitness screen: defect under Tl stress
K03316	*nhaP*	Tl	Unknown	T2F	Fitness screen: defect under Al,Tl stress
K03328	*TC.PST*	Tl	Unknown	T2F	Fitness screen: defect under Co,Ni,Tl stress
K03439	*trmB*	Tl	Unknown	T2F	Fitness screen: defect under Tl stress
K03442	*mscS*	Tl	Unknown	T2F	Fitness screen: defect under Co,Tl stress
K03453	*TC.BASS*	Tl	Unknown	T2F	Fitness screen: defect under Tl stress
K03568	*tldD*	Tl	Unknown	T2F	Fitness screen: defect under Ni,Tl stress
K03593	*mrp*	Tl	Unknown	T2F	Fitness screen: defect under Tl stress
K03701	*uvrA*	Tl	Unknown	T2F	Fitness screen: defect under Co,Ni,Tl,Zn stress
K03885	*ndh*	Tl	Unknown	T2F	Fitness screen: defect under Al,Tl stress
K05368	*fre*	Tl	Unknown	T2F	Fitness screen: defect under Cu,Tl stress
K05592	*deaD*	Tl	Unknown	T2F	Fitness screen: defect under Cu,Tl stress
K05799	*pdhR*	Tl	Sensing/Regulation	T2F	Sensing/regulatory response to Tl,Zn
K05810	*LACC1*	Tl	Unknown	T2F	Fitness screen: defect under Al,Tl stress
K06179	*rluC*	Tl	Unknown	T2F	Fitness screen: defect under Ni,Tl stress
K06941	*rlmN*	Tl	Unknown	T2F	Fitness screen: defect under Cu,Tl stress
K07058	—	Tl	Unknown	T2F	Fitness screen: defect under Co,Ni,Tl stress
K07165	*fecR*	Tl	Unknown	T2F	Fitness screen: defect under Al,Bi,Tl stress
K07263	*pqqL*	Tl	Transport/Homeostasis	T2F	Metal transport/homeostasis (Zn,Co,Tl)
K07390	*grxD*	Tl	Unknown	T2F	Fitness screen: defect under Tl,Zn stress
K08300	*rne*	Tl	Unknown	T2F	Fitness screen: defect under Mn,Tl stress

continued on next page

Table S4 continued.

KO	Gene	Metal	Subcategory	Tier	Biochemical rationale
K08301	*rng*	Tl	Unknown	T2F	Fitness screen: defect under Mn,Tl stress
K09771	*TC.SMR3*	Tl	Resistance/Detox.	T2F	Resistance/detoxification; fitness defect under Tl stress
K11645	*fbaB*	Tl	Unknown	T2F	Fitness screen: defect under Ni,Tl stress
K13628	*iscA*	Tl	Sensing/Regulation	T2F	Sensing/regulatory response to Fe,Tl,Zn
K18707	—	Tl	Unknown	T2F	Fitness screen: defect under Cu,Tl stress
K20534	*gtrB*	Tl	Unknown	T2F	Fitness screen: defect under Al,Tl stress

Table S5. Pairwise overlap of KO sets across the nine metal-specific analyses. Diagonal entries give the total KO count for each metal; off-diagonal entries give the number of KOs shared between the two metal-specific sets. Substantial overlap (e.g. Co–Ni: 43 shared KOs; Ni–Cu: 18; Co–Zn: 17; Fe–Mn: 15) indicates that the nine per-metal PGLS results are not statistically independent. Generated from `data/curated_mrg_ko_ids_v2.csv`; see `data/metal_ko_overlap_matrix.csv`.

	Tl	Fe	Ni	Zn	Al	Co	S	Cu	Mn
Tl	45	3	9	9	10	12	0	9	2
Fe	3	98	7	11	1	8	3	5	15
Ni	9	7	84	13	14	43	0	18	1
Zn	9	11	13	67	4	17	0	12	13
Al	10	1	14	4	34	5	0	11	1
Co	12	8	43	17	5	101	1	10	0
S	0	3	0	0	0	1	74	1	0
Cu	9	5	18	12	11	10	1	71	3
Mn	2	15	1	13	1	0	0	3	34

Supplementary Figures

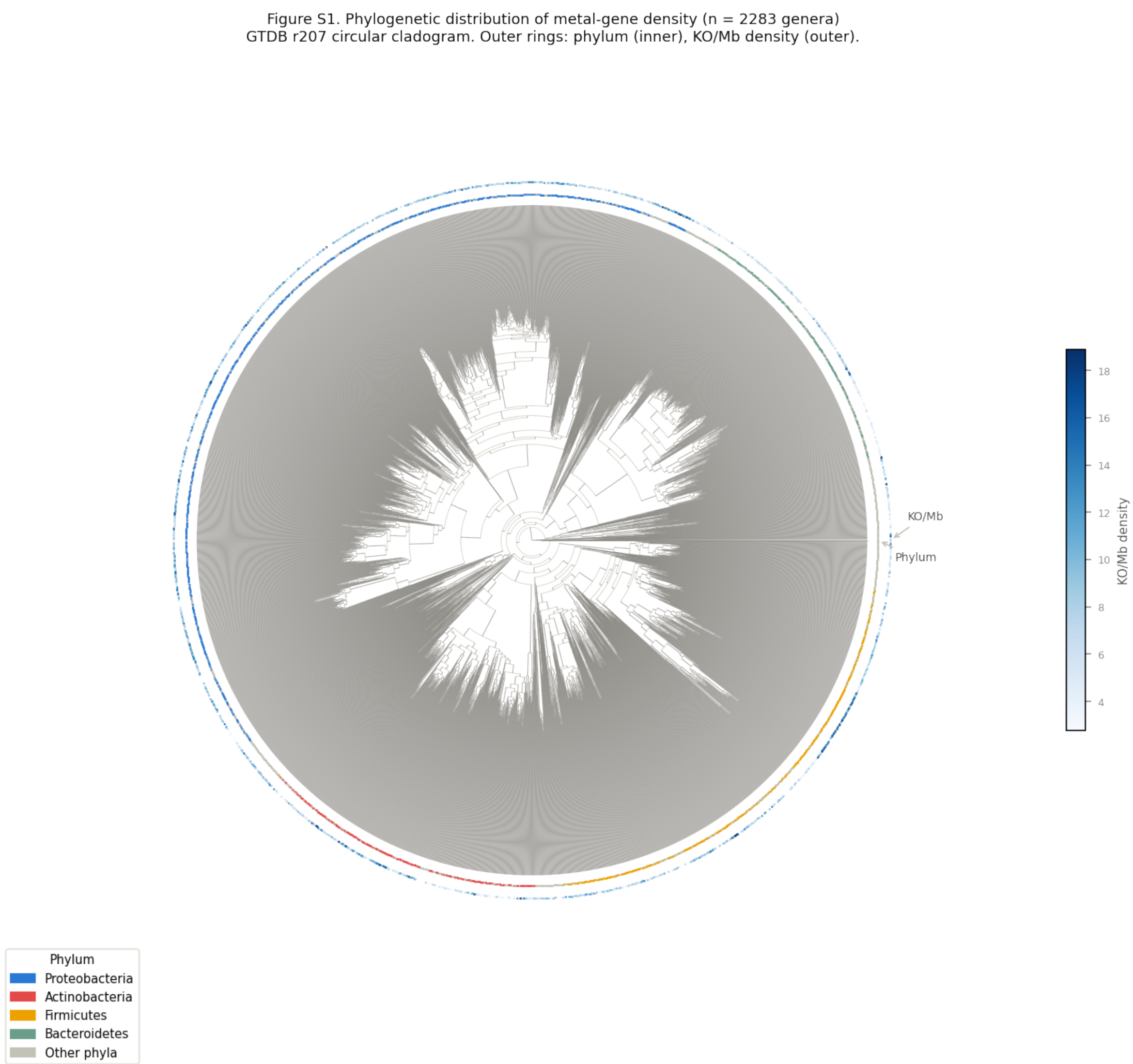


Figure S1. Phylogenetic distribution of genera in the primary dataset. Circular phylogenetic cladogram (GTDB r214v1; $n = 2{,}283$ bacterial genera before pruning; $n = 1{,}574$ with PGLS data, shown with filled tips). Inner ring: Tier 1+2 per-Mb KO density (red–blue gradient; red = high, blue = low). Outer ring: standardised Levins' niche breadth B_{std} (light = specialist, dark = generalist). Arc labels indicate the four focal phyla (Proteobacteria, Actinobacteria, Firmicutes, Bacteroidetes).

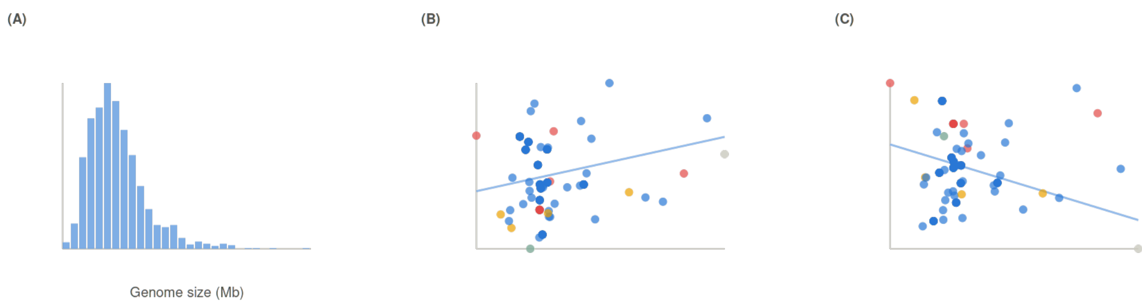


Figure S2. Genome-size distribution and its relationship to niche breadth. (A) Histogram of mean genome size (Mb) across 1,574 genera in the primary dataset. (B) Scatter plot of mean genome size vs standardised niche breadth (B_{std}) with PGLS regression line. (C) Scatter plot of mean genome size vs `ko_per_mb_primary_z`, demonstrating that per-Mb normalisation partially but not fully accounts for genome-size effects. Genome size attenuates the primary β by 46.7%, below the pre-specified 50% confounding threshold.

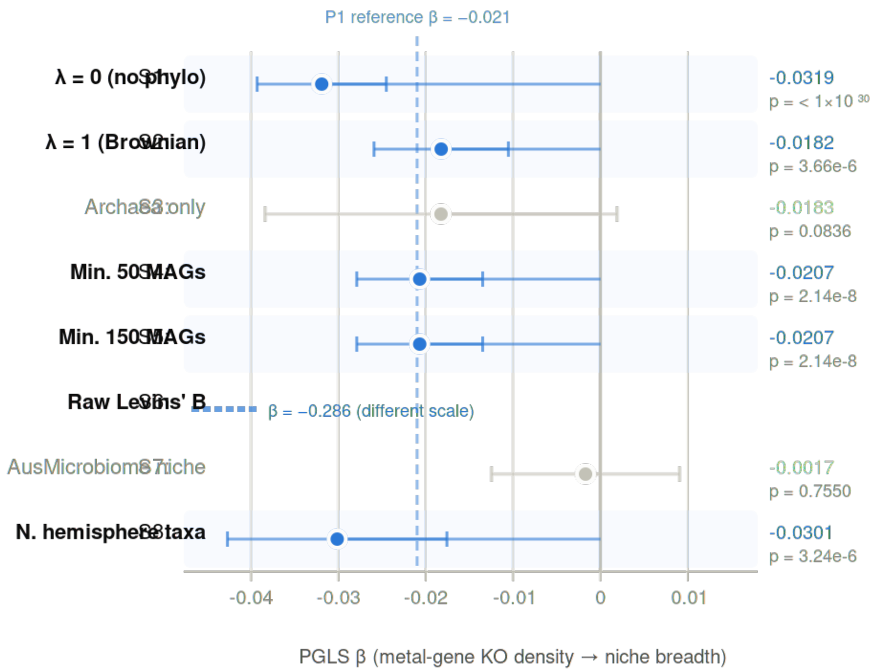


Figure S3. Sensitivity analyses: PGLS β is consistent across eight pre-specified checks. Forest plot of β (with $\pm$1.96 SE error bars) from the primary analysis (bacteria, $n = 1{,}574$) and eight sensitivity analyses: OLS ($\lambda = 0$), Brownian motion ($\lambda = 1$), archaea ($n = 41$, $n_{\min} = 5$), genus-count thresholds ($n_{\min} \geq 50$ and ≥ 150), raw (unstandardised) Levins B, Australia-only subset, northern hemisphere only, and soil-restricted. Filled points: $p < 0.05$; open points: $p \geq 0.05$. Six of eight analyses are significant and directionally consistent.

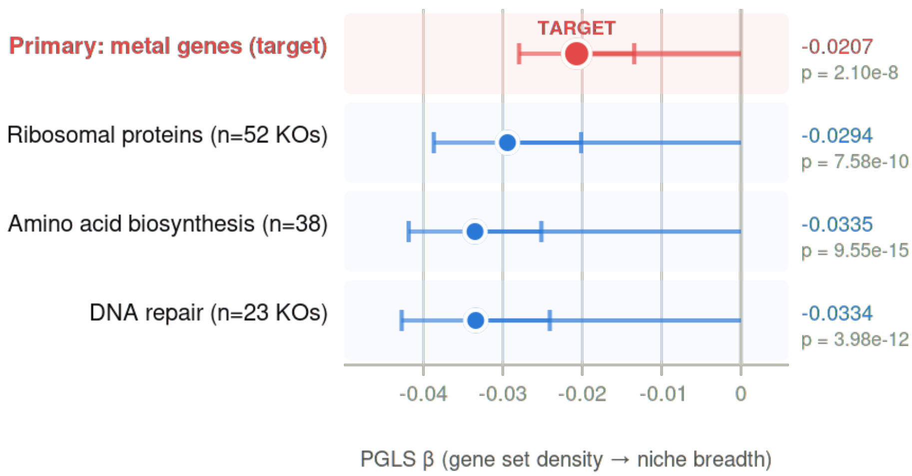


Figure S4. Named negative controls relative to the primary metal-gene estimate. Forest plot of PGLS β for three pre-specified named negative controls (ribosomal proteins, amino acid biosynthesis, DNA repair) alongside the primary metal-gene estimate. All three negative controls are more negative than the primary estimate, confirming that the housekeeping baseline lies at $\beta \approx -0.030$ to -0.035, and the metal-gene $\beta = -0.021$ is weaker than the streamlining baseline. Error bars: ± 1.96 SE.

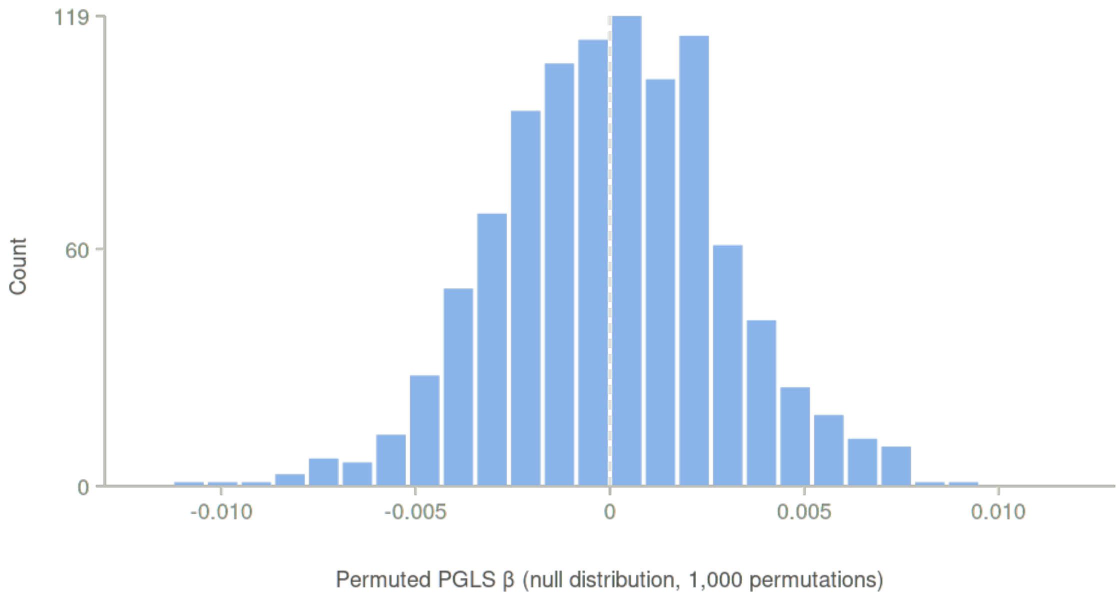


Figure S5. Coreness-matched permutation: the primary β is not distinctive relative to matched gene sets. Histogram of β from 1,000 coreness-matched null KO sets ($n = 140$ KOs each, matched to the primary set by pan-genome prevalence decile). Red vertical line marks the observed $\beta = -0.021$ (empirical $p = 0.298$). Null distribution: median $= -0.018$, SD $= 0.007$. The primary metal-gene β falls within the null distribution, indicating that the streamlining signal is not statistically unusual for a gene set of this size and prevalence distribution.

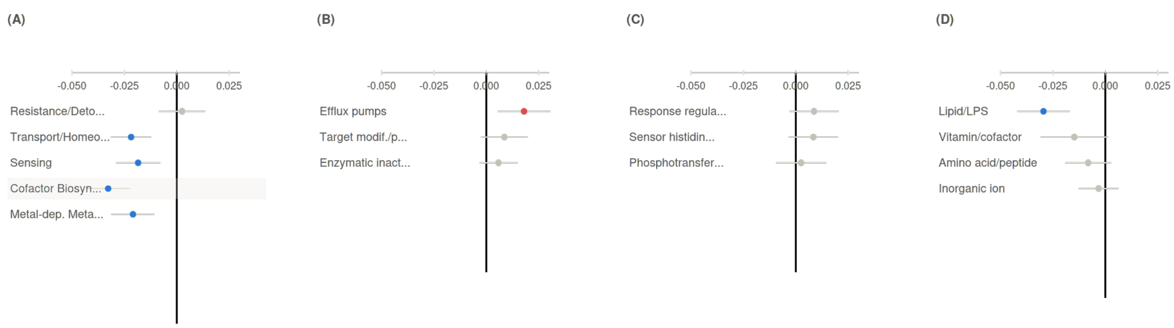


Figure S6. Functional subcategory comparison across AMR, TCS, and ABC transporter families. Side-by-side forest plots of PGLS β for subcategories within four gene families: metal genes (five functional subcategories within the 140-KO primary set), AMR (efflux pumps, enzymatic inactivation, target modification), TCS (sensor kinases, response regulators, phosphotransfer proteins), and ABC transporters (by substrate class). Demonstrates that the resistance-null / cofactor-significant split is specific to the metal-gene set; AMR subcategories are all positive (anti-streamlining), TCS subcategories are all positive, while ABC transporter Lipid/LPS subcategory shows the only negative signal ($\beta = -0.030$, $q = 3.3 \times 10^{-5}$). Error bars: ± 1.96 SE; filled points: BH-FDR $q < 0.05$.

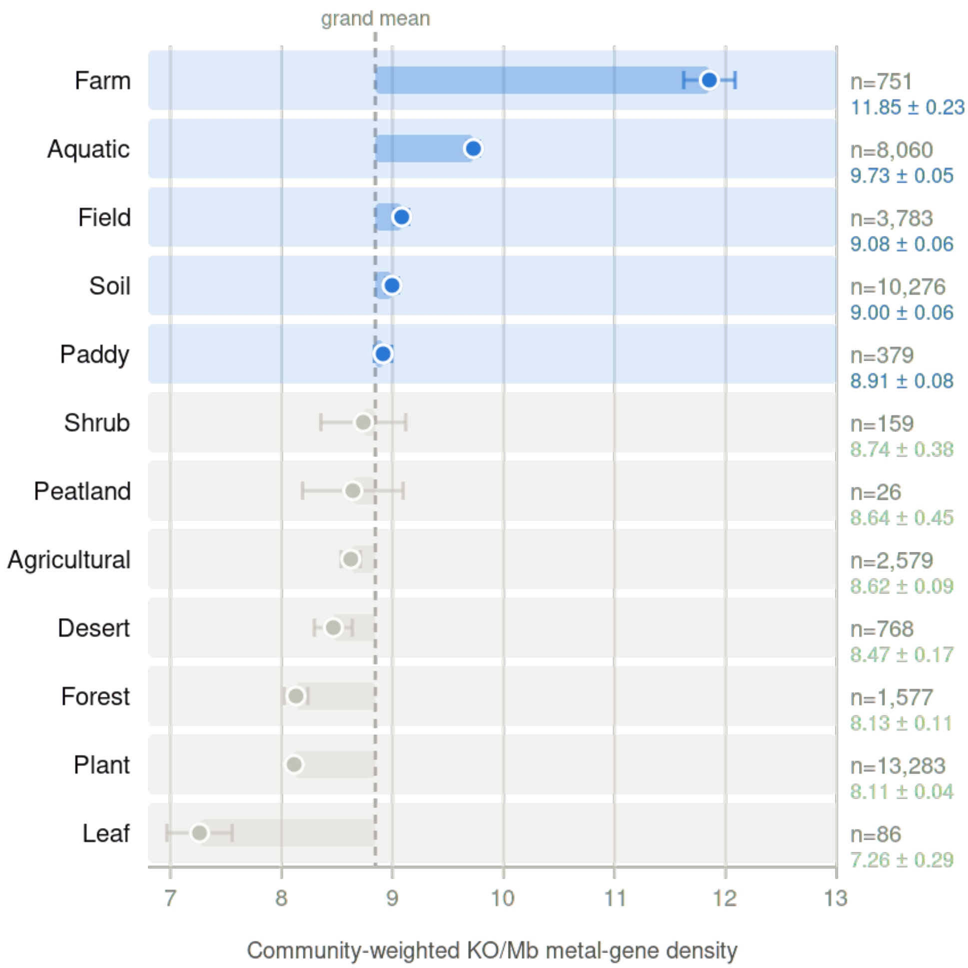


Figure S7. Community-weighted metal-gene investment across biomes. Bar plot of mean `ko_per_mb_primary_z` by MicrobeAtlas `Env_Level_1` habitat category (n = 12 categories with > 100 samples), weighted by genus relative abundance within each habitat. Error bars: ±SE. Marine habitats show consistently lower community-weighted metal-gene density than terrestrial or host-associated environments, consistent with selective pressure from oligotrophic marine conditions favouring extreme genome streamlining.

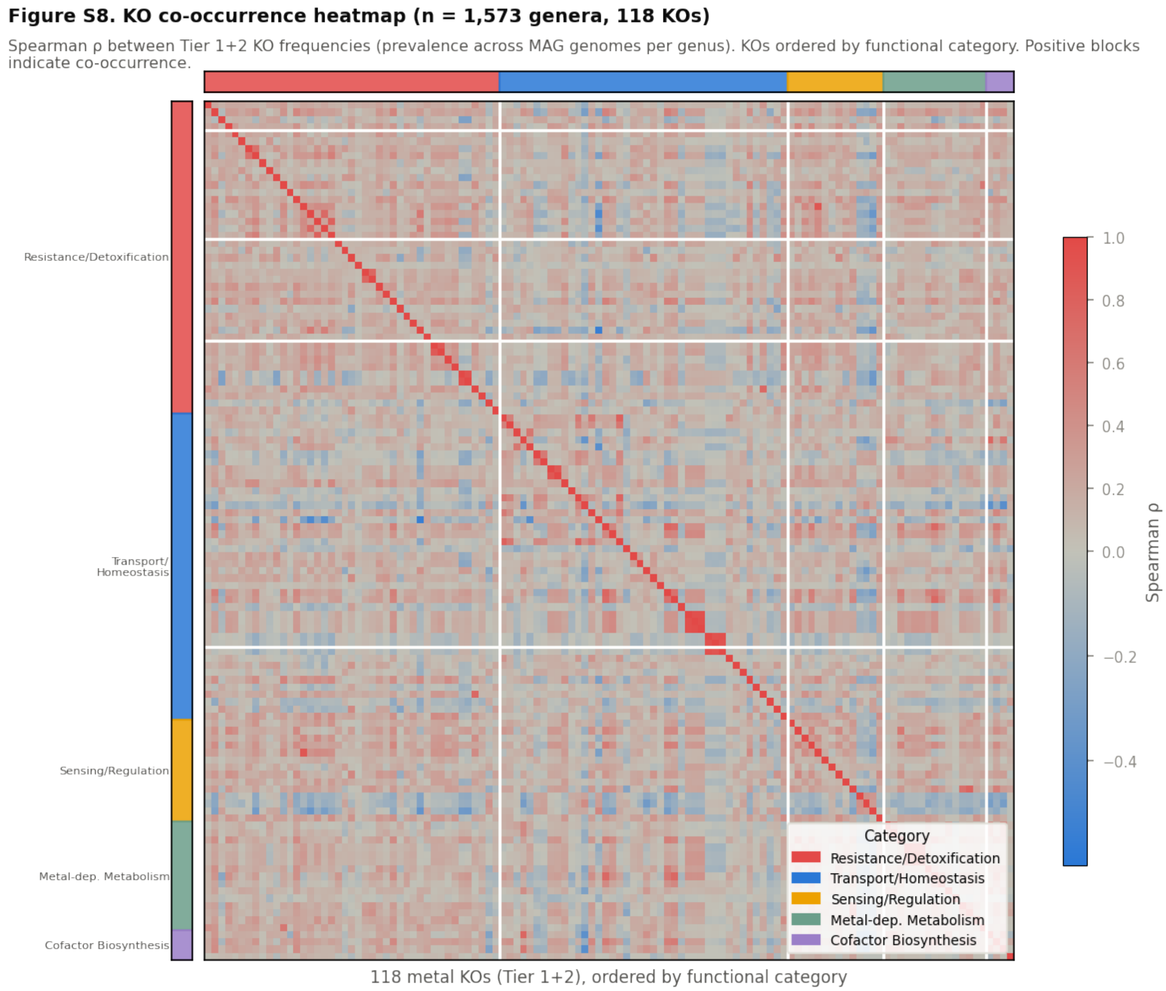


Figure S8. Tier co-occurrence heatmap. Spearman ρ heatmap of cross-genus co-occurrence between individual Tier 1 and Tier 2 KO densities at the genus level (118 × 118 KOs represented; BH-FDR $q < 0.05$ cells highlighted). KOs are clustered by functional subcategory (indicated by coloured column/row bars). Demonstrates the within-panel correlation structure: transport KOs form a broad positive cluster; resistance KOs and cofactor KOs form distinct modules weakly correlated with the transport block, justifying their treatment as distinct analytical subcategories.

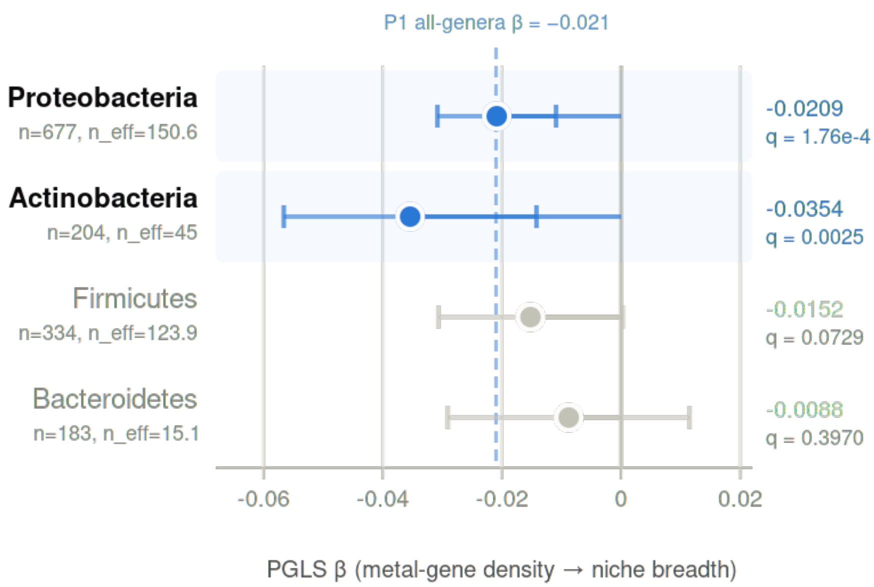


Figure S9. Clade-stratified PGLS: signal within Proteobacteria and Actinobacteria. Forest plot of PGLS β for four phylum-stratified analyses (Proteobacteria, Actinobacteria, Firmicutes, Bacteroidetes) alongside the primary all-bacteria estimate. Error bars: ± 1.96 SE; filled points: BH-FDR $q < 0.05$ across the four phyla. Cochran's $Q = 3.60$ (df $= 3$, $p = 0.309$; $I^2 = 16.6\%$): no significant heterogeneity across phyla. The signal is statistically significant in Proteobacteria and Actinobacteria; Firmicutes and Bacteroidetes are directionally consistent but underpowered.

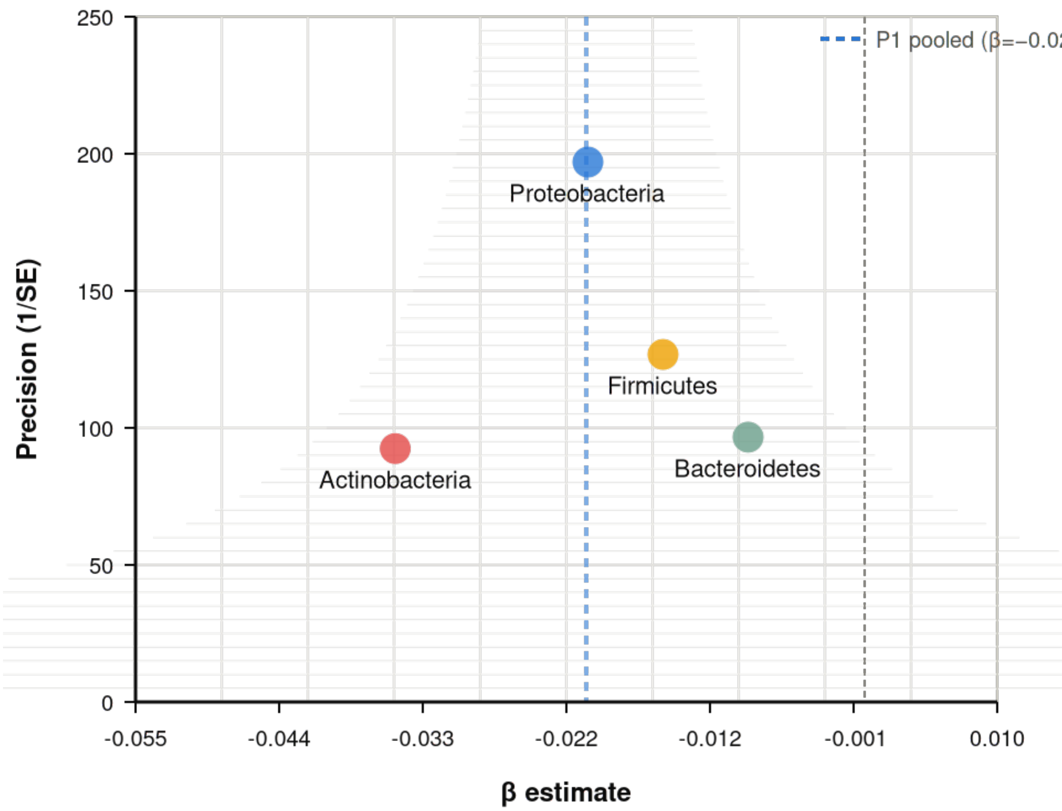


Figure S10. Heterogeneity test: phylum-level β estimates are statistically consistent. Funnel plot of phylum-specific β (x-axis) vs precision 1/SE (y-axis) for the four clade-stratified estimates (Proteobacteria, Actinobacteria, Firmicutes, Bacteroidetes). Cochran's $Q = 3.60$ (df $= 3$, $p = 0.309$; $I^2 = 16.6\%$): no significant heterogeneity; the primary effect is not concentrated in a single clade.

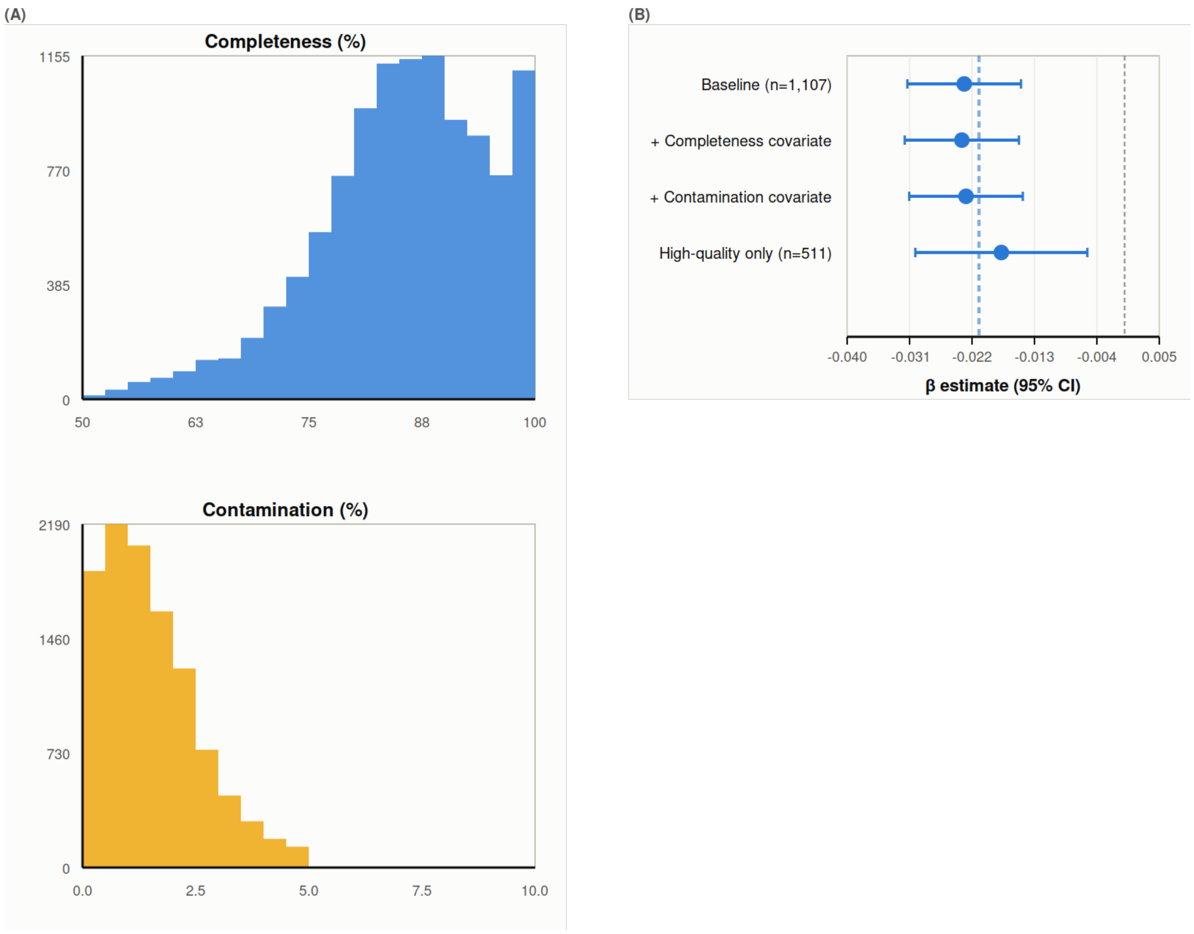


Figure S11. MAG quality metrics and their effect on the primary association. (A) Histograms of mean per-genus completeness (%) and contamination (%) for the $n = 1{,}107$ genera with CheckM quality metadata. (B) Forest plot of PGLS β from four models: primary analysis, primary with completeness added as covariate, primary with contamination as covariate, and primary restricted to high-quality MAGs (completeness $\geq 90\%$, contamination $\leq 5\%$; $n = 511$ genera). Error bars: ± 1.96 SE. The primary β is robust to MAG quality variation: adding quality covariates leaves β unchanged at -0.023 ($p \approx 3 \times 10^{-8}$).

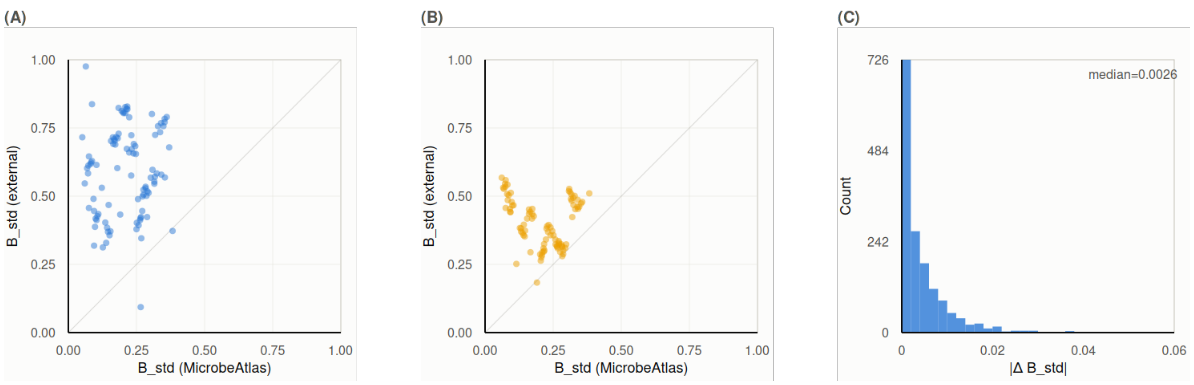


Figure S12. Niche-breadth metric validation: B_{std} correlates with EMP and MGnify biome metrics. (A) Scatter plot of MicrobeAtlas B_{std} vs EMP-derived Levins' B_{std} for 539 overlapping genera (Spearman $\rho = 0.211$, $p = 7.7 \times 10^{-7}$). (B) Scatter plot of MicrobeAtlas B_{std} vs MGnify biome Shannon diversity per genus ($\rho = +0.063$, $p = 0.046$). (C) Bootstrap stability: histogram of $|\Delta B_{\mathrm{std}}|$ between the original and mean bootstrap estimate across 100 resamples ($n = 1{,}574$ genera). The low median $|\Delta|$ confirms that B_{std} is stable at the current MicrobeAtlas sample size.

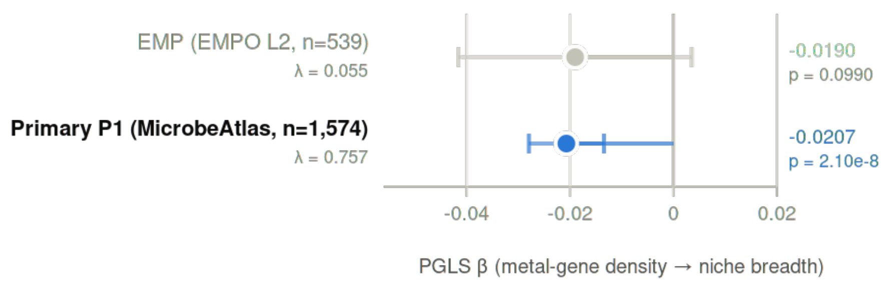


Figure S13. EMP PGLS: directionally consistent at $n = 539$. Scatter plot of EMP-derived Levins' B_{std} vs `ko_per_mb_primary_z` for 539 genera shared between the Earth Microbiome Project and the primary pangenome dataset. PGLS regression line (blue): $\beta = -0.019$, $p = 0.099$ (non-significant; directionally consistent with the primary result). Primary MicrobeAtlas regression line (grey) overlaid for comparison. The lower significance reflects the smaller sample size ($n = 539$ vs 1,574) and the coarser EMPO-2 habitat categorisation used as the niche axis.

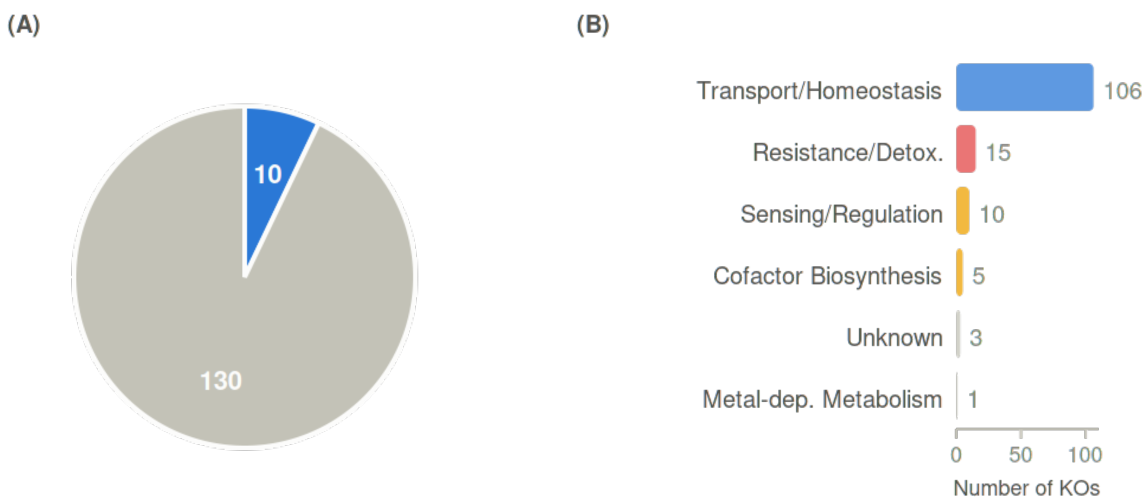


Figure S14. Pfam / InterPro audit of 140 primary KOs. Left: pie chart showing classification of 140 KOs by Pfam domain outcome: metal-binding clan confirmed (10/140, 7.1%; five InterPro clans: CL0704, CL0344, CL0486, CL0361, CL0193); metal-processing Pfam but no metal-binding clan (113/140; e.g., ABC transporter scaffolds PF00005/PF00950, MerR-family sensors); no Pfam annotation (2/140); no representative bacterial gene in KEGG REST (15/140). Right: breakdown by functional subcategory, demonstrating that low clan coverage is expected for transporters (which act via binding-pocket selectivity) but not for cofactor enzymes (which carry structural metal-binding domains).

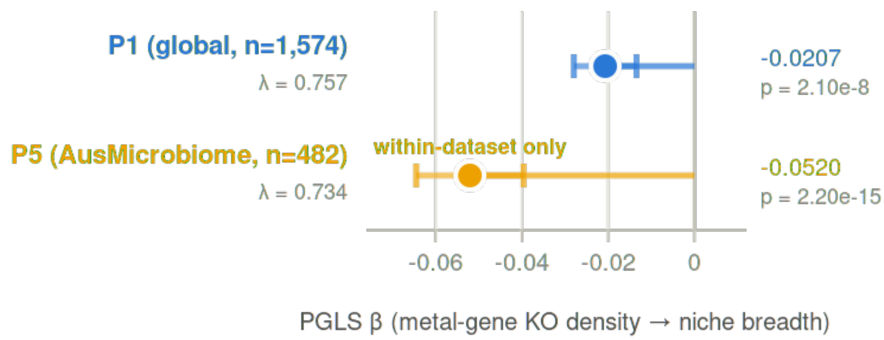


Figure S15. AusMicrobiome within-dataset PGLS and composition diagnostics. (A) Scatter plot of B_{std} vs `ko_per_mb_primary_z` for 482 Australian genera ($\beta = -0.052$, $p = 2.2 \times 10^{-15}$). (B) Phylum composition bar chart comparing the global (n = 1,574) and Australian (n = 482) genus panels; Proteobacteria are enriched and Firmicutes are depleted in the continental subset. (C) Primary analysis restricted to the same 482 genera: $\beta = -0.052$, confirming that the larger within-dataset β reflects a composition effect (the 482 Australian genera, dominated by Proteobacteria, collectively lie in a steeper part of the global regression).

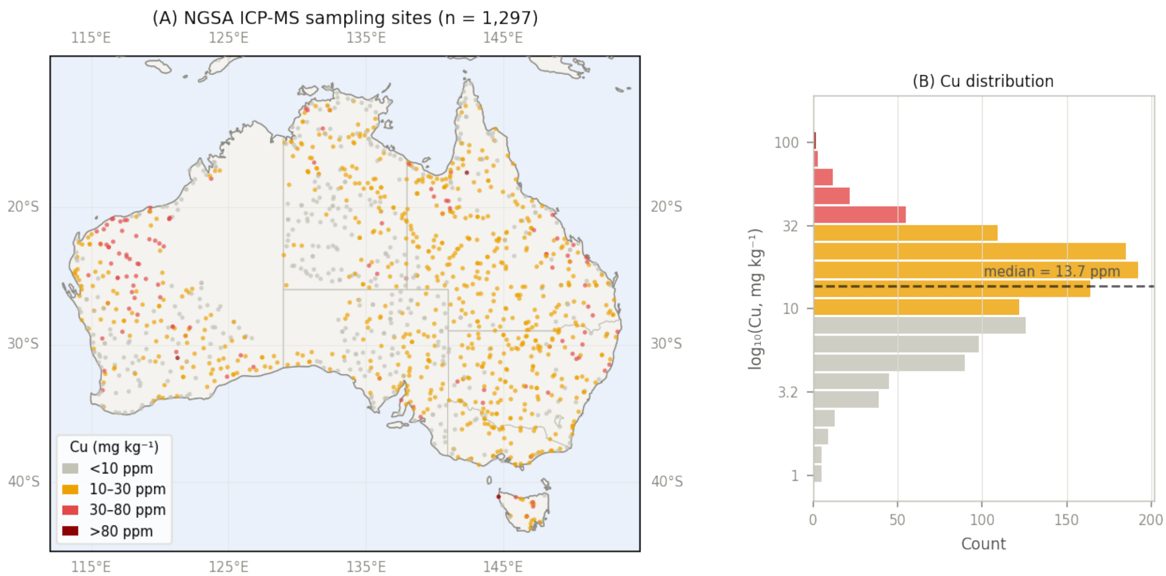
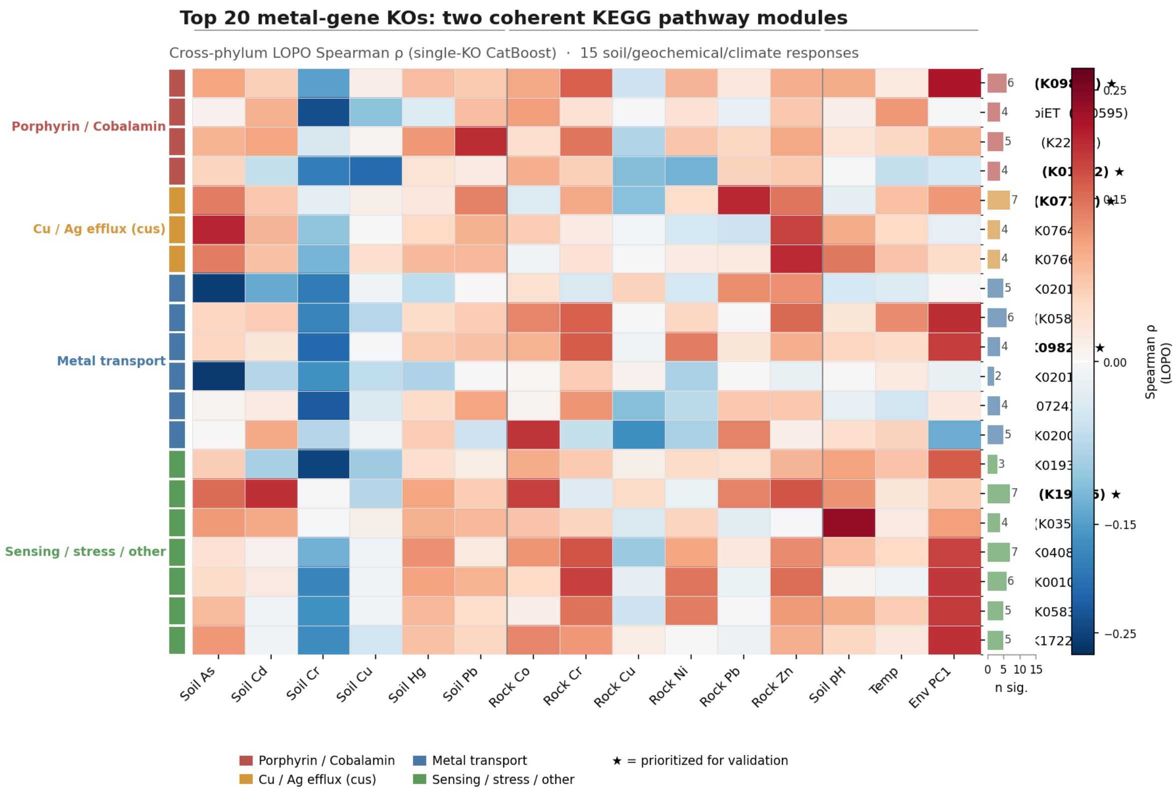


Figure S16. Geochemical replication: spatial distribution of NGSA sampling sites and genus-level metal concentration assignments. Map of Australia showing National Geochemical Survey of Australia (NGSA) ICP-MS soil sampling sites ($n \approx 1{,}200$ sites) and the 200 km radius used for spatial assignment of soil metal concentrations to AusMicrobiome sampling locations. Inset: histogram of per-genus mean Cu concentration (ICP-MS, $\mathrm{mg\,kg^{-1}}$) for the 482 genera with NGSA coverage. Sites are distributed across all major geological provinces, providing broad coverage of Australian bedrock diversity. The 200 km radius was chosen to maximise genus coverage while remaining within a geologically coherent region for most sites.



Top 20 metal-gene KOs: two coherent KEGG pathway modules
Cross-phylum LOPO Spearman ρ (single-KO CatBoost) · 15 soil/geochemical/climate responses
Porphyrin / Cobalamin
Cu / Ag efflux (cus)
Metal transport
Sensing / stress / other
Soil As
Soil Cd
Soil Cr
Soil Cu
Soil Hg
Soil Pb
Rock Co
Rock Cr
Rock Cu
Rock Ni
Rock Pb
Rock Zn
Soil pH
Temp
Env PC1
n sig.
Spearman ρ (LOPO)
0.25
0.15
0.00
−0.15
−0.25
Porphyrin / Cobalamin
Cu / Ag efflux (cus)
Metal transport
Sensing / stress / other
★ = prioritized for validation

References

[1] Pal C, Bengtsson-Palme J, Rensing C, Kristiansson E, Larsson DGJ (2014) BacMet: antibacterial biocide and metal resistance genes database. *Nucleic Acids Res* **42**:D737–D743. doi:10.1093/nar/gkt1252

[2] Price MN, Wetmore KM, Waters RJ, *et al.* (2018) Mutant phenotypes for thousands of bacterial genes of unknown function. *Nature* **557**:503–509. doi:10.1038/s41586-018-0124-0

[3] Levins R (1968) *Evolution in Changing Environments.* Princeton University Press, Princeton NJ.

[4] Colwell RK, Futuyma DJ (1971) On the measurement of niche breadth and overlap. *Ecology* **52**:567–576. doi:10.2307/1934144

[5] Sunagawa S, Coelho LP, Chaffron S, Kultima JR, Labadie K, Salazar G *et al.* (2015) Structure and function of the global ocean microbiome. *Science* **348**:1261359. doi:10.1126/science.1261359

[6] Arkin AP, Cottingham RW, Henry CS, *et al.* (2018) KBase: The United States Department of Energy Systems Biology Knowledgebase. *Nat Biotechnol* **36**:566–569. doi:10.1038/nbt.4163

[7] Parks DH, Chuvochina M, Rinke C, Mussig AJ, Chaumeil P-A, Hugenholtz P (2022) GTDB: an ongoing census of bacterial and archaeal diversity through a phylogenetically consistent, rank-normalised and complete genome-based taxonomy. *Nucleic Acids Res* **50**:D785–D794. doi:10.1093/nar/gkab776

[8] Virtanen P, Gommers R, Oliphant TE, *et al.* (2020) SciPy 1.0: Fundamental algorithms for scientific computing in Python. *Nat Methods* **17**:261–272. doi:10.1038/s41592-019-0686-2

[9] Prokhorenkova LO, Gusev G, Vorobev A, Dorogush AV, Gulin A (2018) CatBoost: unbiased boosting with categorical features. In: *Advances in Neural Information Processing Systems* **31**, pp 6638–6648.

[10] Lundberg SM, Lee S-I (2017) A unified approach to interpreting model predictions. In: *Advances in Neural Information Processing Systems* **30**.

[11] Kanehisa M, Furumichi M, Sato Y, Kawashima M, Ishiguro-Watanabe M (2023) KEGG for taxonomy-based analysis of pathways and genomes. *Nucleic Acids Res* **51**:D587–D592. doi:10.1093/nar/gkac963

[12] Qi C, Hu T, Zheng Y, Wu M, Tang FHM, Liu M, *et al.* (2025) Global and regional patterns of soil metal(loid) mobility and associated risks. *Nat Commun* **16**:2947. doi:10.1038/s41467-025-58026-8

[13] von Meijenfeldt FAB, Hogeweg P, Dutilh BE (2023) A social niche breadth score reveals niche range strategies of generalists and specialists. *Nat Ecol Evol* **7**:768–781. doi:10.1038/s41559-023-02027-7

[14] Pagel M (1999) Inferring the historical patterns of biological evolution. *Nature* **401**:877–884.

[15] Grafen A (1989) The phylogenetic regression. *Philos Trans R Soc Lond B Biol Sci* **326**:119–157.

[16] Adato O, Ninyo N, Gophna U, Snir S (2015) Detecting horizontal gene transfer between closely related taxa. *PLoS Comput Biol* **11**:e1004408. doi:10.1371/journal.pcbi.1004408

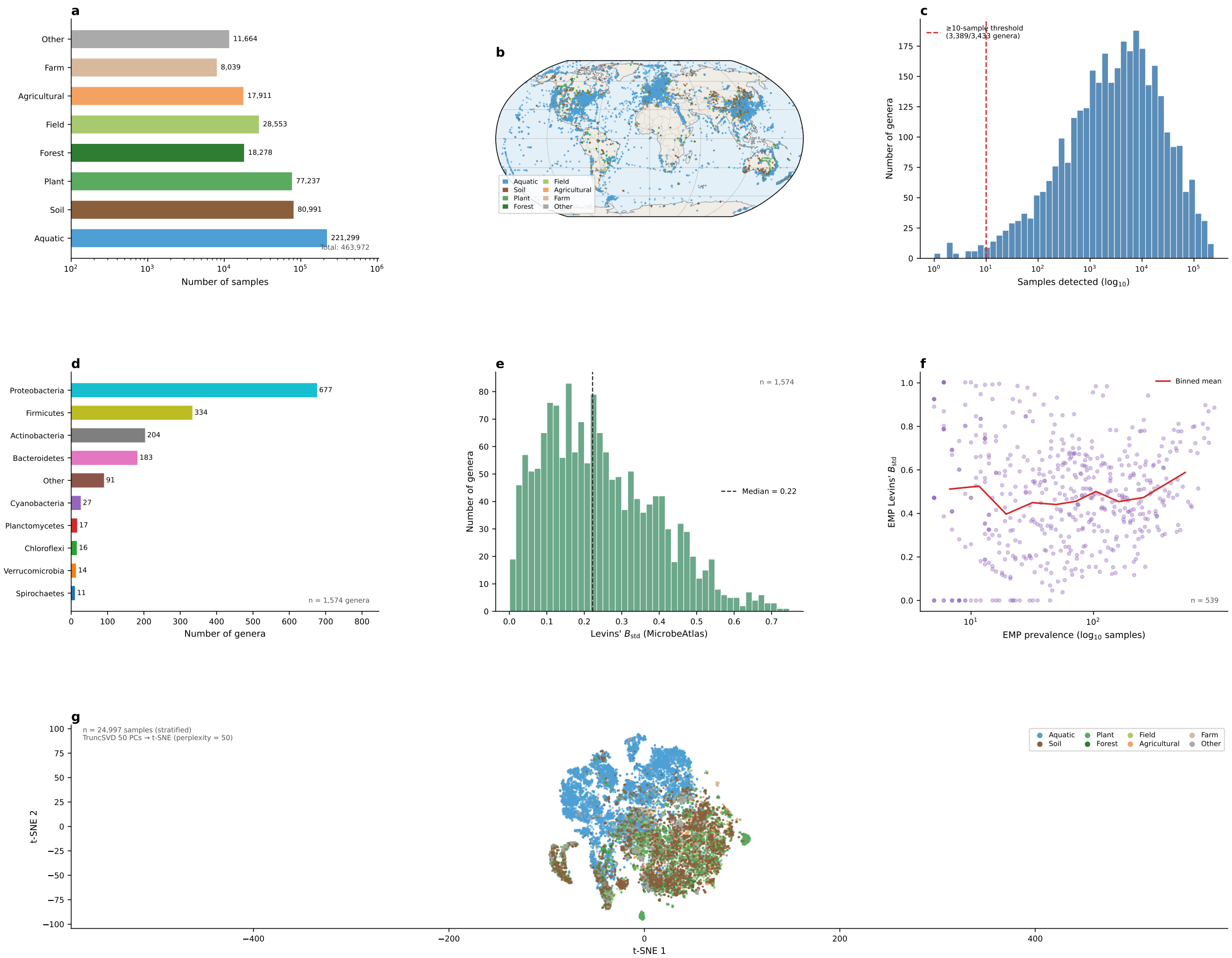


Figure S18. Overview of the MicrobeAtlas dataset used in this study. (a) Sample counts per biome category ($\log_{10}$ scale; N = 463,972 total samples). **(b)** Global distribution of 386,064 geolocalised samples, coloured by biome; Robinson projection. **(c)** Prevalence distribution of 3,433 unique genera across all MicrobeAtlas samples ($\log_{10}$ scale); the dashed line marks the $\geq$ 10-sample inclusion threshold used throughout the analysis (3,389 of 3,433 genera pass). **(d)** Phylum composition of the 1,574 genera in the primary PGLS dataset; top 9 phyla shown individually, remainder grouped as Other. **(e)** Distribution of standardised Levins' niche breadth (B_{std}; MicrobeAtlas) across the 1,574 primary genera; dashed line indicates the median. **(f)** EMP prevalence vs. EMP B_{std} for the 539 genera in the EMP validation dataset (Spearman correlation analysis); the red line is the binned mean across 10 equal-frequency bins. **(g)** t-SNE of 24,997 stratified MicrobeAtlas samples (n = 25,000 target; proportional to biome count), coloured by biome. Genus-level binary presence/absence vectors (3,389 genera) were reduced to 50 principal components via Truncated SVD (cumulative explained variance = 0.438) before t-SNE (sklearn; perplexity = 50, max_iter = 1,000, `init=pca`). Each point is one sample; structure reflects shared genera-level community composition. Data sources: `data/sample_latlon_env.csv` (panels b,c), `data/genus_microbeatlas_sample_counts.csv` (panel c), `data/01_pgls_input_bacteria.csv` (panels d,e), `data/emp_niche_pgls_input.csv` (panel f), `data/tsne_embedding.csv` (panel g); scripts: `scripts/make_figS1_sample_overview.py`, `scripts/compute_tsne_embedding.py`.

[17] Tonkin-Hill G, Gladstone RA, Pöntinen AK, Arredondo-Alonso S, Bentley SD, Corander J (2023) Robust analysis of prokaryotic pangenome gene gain and loss rates with Panstripe. *Genome Res* **33**:129–140. doi:10.1101/gr.277340.122